بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

LIVE ONLINE COURSE | COHORT FALL 2026

AI & Machine Learning for Civil Engineers

SESSION 03 · LINEAR REGRESSION, ERROR METRICS & CROSS-VALIDATION

Your First Engineering Prediction Model

Can a model learn a useful relationship from the data we prepared, and how would we know?

DATA → MODEL → PREDICTION → EVALUATION → DECISION



Dr. Abdul Waris
Research Associate · Universität Siegen, Germany

Technology partner QUDELTA STUDIOS

WHERE ARE WE? · FROM DATA PREPARATION TO PREDICTION

We have prepared the data. Now we predict.

SESSIONS 01–02: WE ALREADY LEARNED TO

Understand an engineering dataset	`data dictionary`
Identify features and target	`X, y`
Check data quality	`isna · duplicated`
Explore relationships	`hist · corr`
Avoid data leakage	`known in time?`
Split data into training and test sets	`train_test_split`

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TODAY · SESSION 03

Can we use these inputs to predict an engineering response?

Build our first model, then **evaluate it honestly.**

THE COURSE AT A GLANCE

Eight sessions. One engineering workflow.

PART 1 · FOUNDATIONS

01 From Engineering Problems to Python — DONE
AI/ML basics, Google Colab, NumPy & pandas

02 From Raw Engineering Data to an ML-Ready Dataset — DONE
EDA, data quality, outliers, train/test split, leakage

03 Your First Engineering Prediction Model — TODAY
Linear regression, error metrics, cross-validation

04 Nonlinear Engineering Problems: Trees & Ensembles
Decision trees, Random Forest, boosting

PART 2 · PREDICTION TO DECISION

05 From Prediction to Engineering Decision: Classification
Logistic regression, confusion matrix, precision & recall

06 Artificial Neural Networks
Architecture, activations, training, overfitting

07 Can We Trust Our ML Model?
Feature importance, SHAP, engineering judgement
Instructor's research area

08 From Prediction to Probability
Monte Carlo, ML surrogates, probability of failure
Instructor's research area

TODAY · SESSION 03

Your first engineering prediction model

TODAY · SESSION 03

Your first engineering prediction model

01

Build the model

The question · a baseline · training · linear regression · the maths inside



TODAY · SESSION 03

Your first engineering prediction model

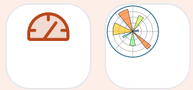
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MAE · RMSE · R^2 · train vs test · parity and residual plots



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03 **Can we trust it?**
Split sensitivity · cross-validation · the test-set rule · engineering judgement

TODAY · SESSION 03

Your first engineering prediction model

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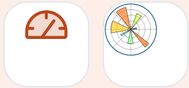
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BY THE END OF TODAY YOU CAN

- Fit a linear-regression model and compare it with a baseline
- Explain what training, a cost function and gradient descent do
- Read MAE, RMSE and R^2, and diagnose a model with residuals
- Use cross-validation for decisions and the test set only once

01

PART 01 · BUILD THE MODEL

From engineering question to a trained model

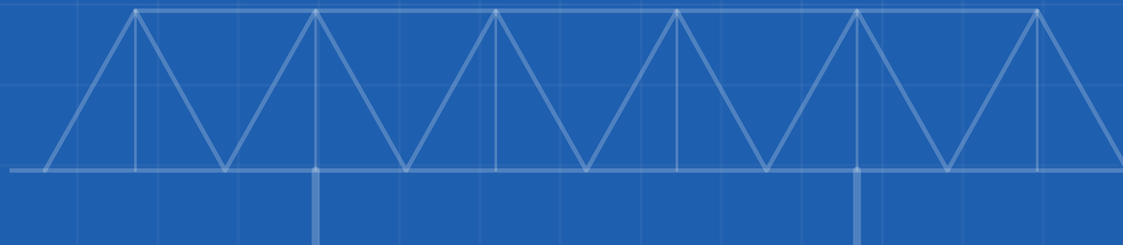
a Our engineering question

b A baseline to beat

c What “training” means

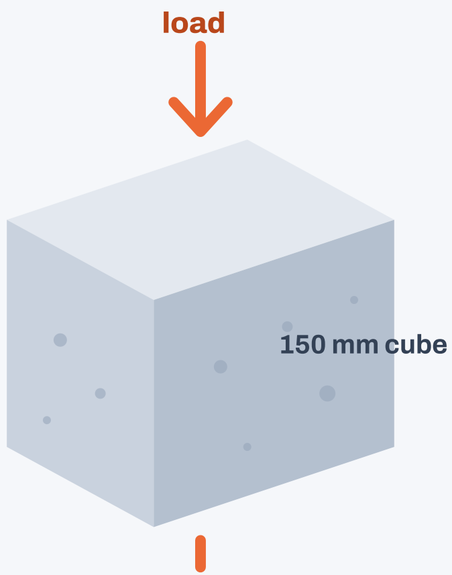
d Linear regression and its weights

e The maths: cost, gradient descent, scaling



OUR ENGINEERING QUESTION

Can we predict concrete compressive strength?

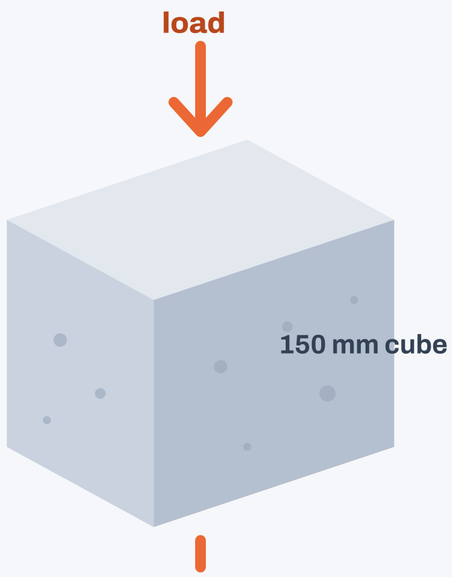


We know the mix and the curing age. Can we estimate the strength **without performing the test?**

Cement kg/m³	**Blast-furnace slag** kg/m³	**Fly ash** kg/m³	**Water** kg/m³
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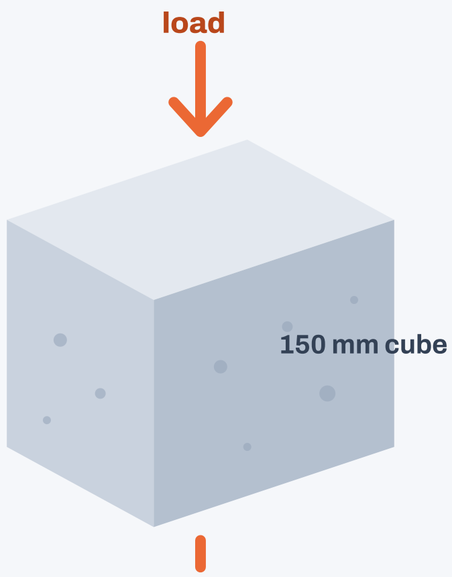
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y **Compressive strength** MPa

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8 INPUTS → MODEL →

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“If I give you a new concrete mix, could you estimate its strength without testing it?”

THE DATASET · UCI CONCRETE COMPRESSIVE STRENGTH

Our first ML prediction dataset

1,030

concrete mixes in the file

Data: UCI Concrete Compressive Strength (Yeh, 1998). From today the notebook uses `df.drop_duplicates()` before splitting, as recommended in Session 02.

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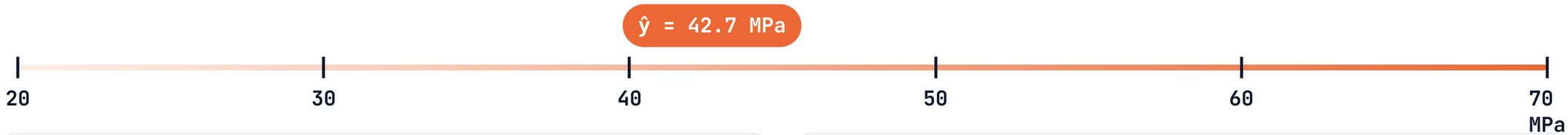
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THE ML PROBLEM

A regression problem: the target is a continuous quantity



Inputs · X **Mix proportions + age** — 8 numbers per mix

Target · y **Compressive strength** — any value on the line, not a category

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REGRESSION IN ONE PICTURE · A REAL TEST MIX

Regression = predicting a number, then checking it

THE MODEL RECEIVES · X

Cement	146.5
Slag	114.6
Fly ash	89.3
Water	201.9
Superplasticizer	8.8
Aggregates (coarse · fine)	860 · 830
Age (days)	28

One of the 201 test mixes in the notebook (inputs in kg/m³). The model never saw it during training.

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Prediction error ŷ – y	+2.5 MPa

Every regression model produces predictions that we must compare against observations.

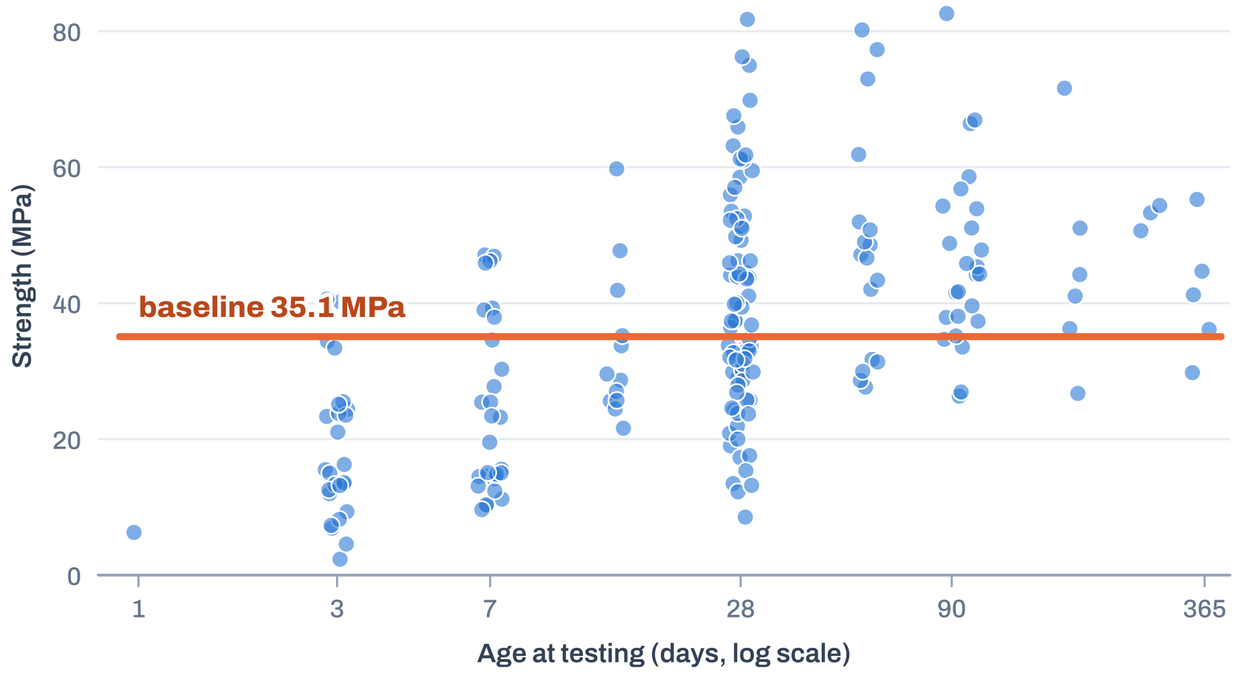
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BEFORE ML: ESTABLISH A BASELINE

Is our model actually useful? Compared with what?

The mean baseline ignores all eight inputs

It predicts the average training strength for every mix. Dots: the 201 test mixes.

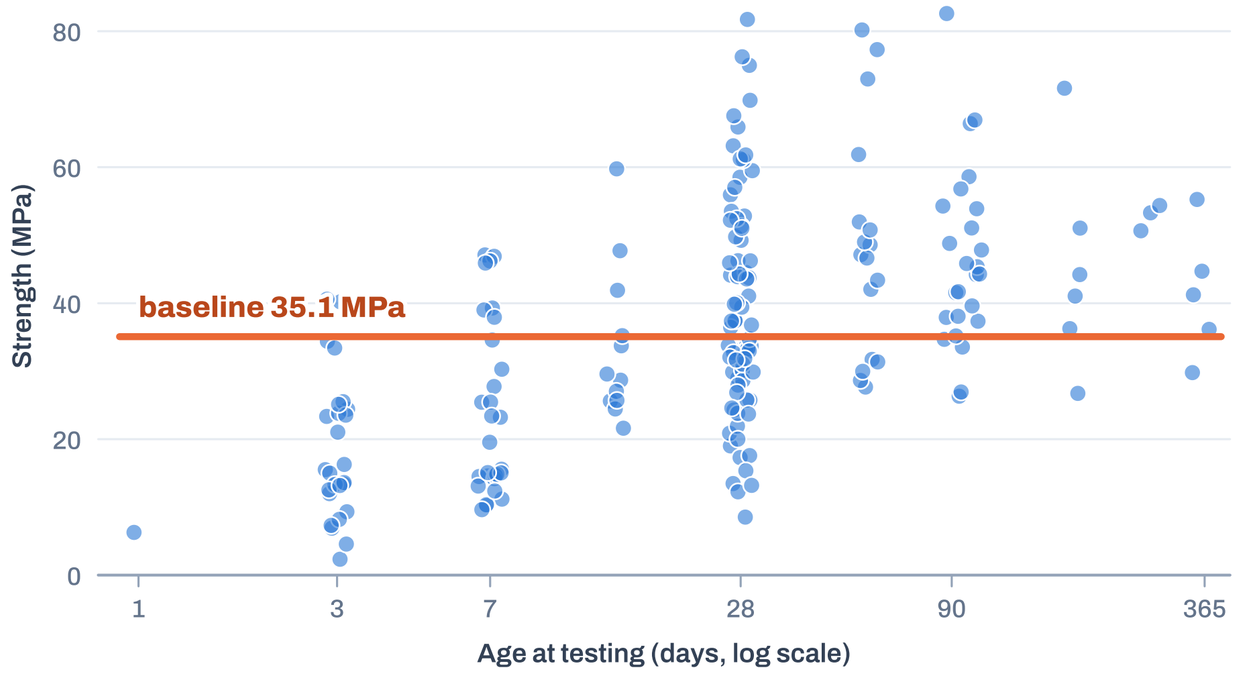


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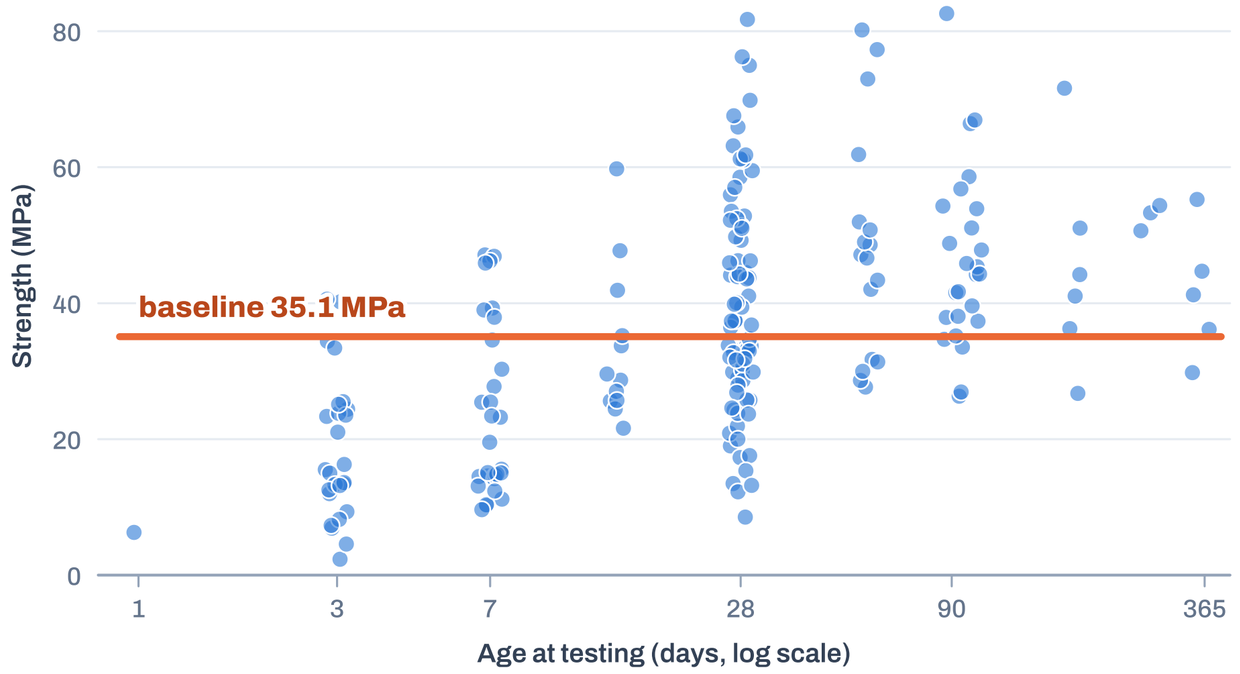
Average training strength = **35.1 MPa** → predicted for every concrete mix.

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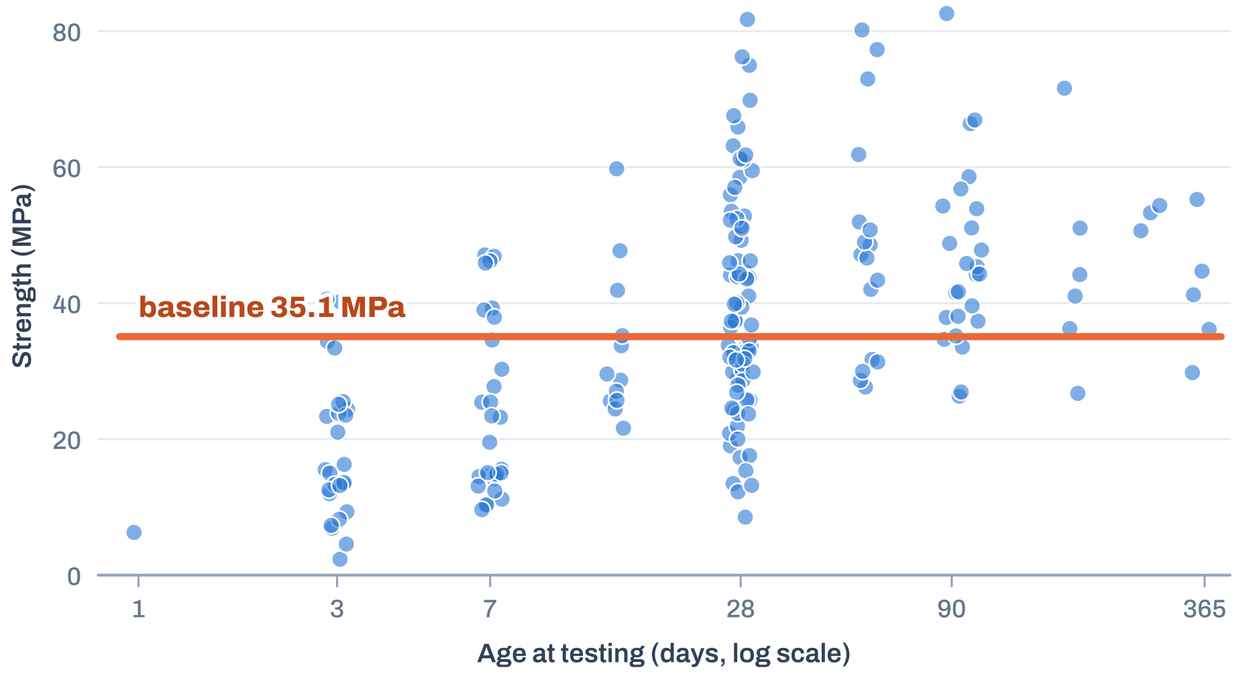
On the test mixes it is off by **14.0 MPa** on average.

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A RESEARCH HABIT

A useful model must beat this.

Never report "R^2 = 0.75" without saying: compared with what?

WHAT DOES “TRAINING” MEAN?

Training = learning model parameters from examples

Training examples
804 mixes with their measured strength

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Session03_first_model.ipynb — fit

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from sklearn.linear_model import LinearRegression
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model.fit(X_train, y_train)
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FOR LINEAR REGRESSION THE PARAMETERS ARE

`intercept_` one number: the starting value

`coef_` one weight for each of the 8 inputs

OUR FIRST PROTECTION AGAINST OVERCONFIDENCE

Why don’t we train on everything?

Studying with the exact questions that will appear in the exam gives a high score that says little about new questions.

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80% · Training data

804 mixes

Fitting the model | Model development | Cross-validation

20% · Test

201 mixes

Final evaluation

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Final evaluation

The test set represents new data from the model’s perspective.

LINEAR REGRESSION: OUR FIRST MODEL

Why start with linear regression?

THE PREDICTION IS A WEIGHTED SUM OF THE INPUTS

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_8 x_8$$

x_j
the input variables

β_j
learned coefficients (weights)

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$\hat{y}$
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Simple does not mean useless.

WHAT DOES THE MODEL LEARN? · OUR FITTED MODEL

The model learns the weights

FITTED ON 804 TRAINING MIXES

$$\textit{Strength} = -23.39 + 0.117 \cdot \text{cement} + 0.098 \cdot \text{slag} + 0.085 \cdot \text{fly ash} - 0.131 \cdot \text{water} + 0.331 \cdot \text{superplasticizer} + 0.016 \cdot \text{coarse} + 0.021 \cdot \text{fine} + 0.111 \cdot \text{age}$$

Units: MPa per kg/m^3 for the mix quantities, MPa per day for age; intercept in MPa. Rounded to three decimals.

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WE DON'T CHOOSE THESE NUMBERS

The algorithm estimates them from the examples we provide. This is where **the data starts writing the model,** instead of us writing the rules.

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Session03_first_model.ipynb — weights

```
model.intercept_    # β0
model.coef_         # β1 … β8
```

OUTPUT

```
-23.39
[0.117, 0.098, 0.085, -0.131, 0.331, 0.016, 0.021,
0.111]
```

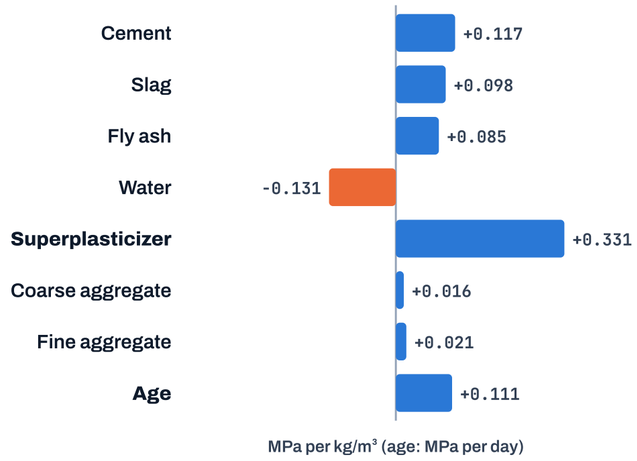
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LOOKING INSIDE THE MODEL

What do the coefficients tell us?

Raw coefficients

Change in predicted strength per unit of each input, others held constant.

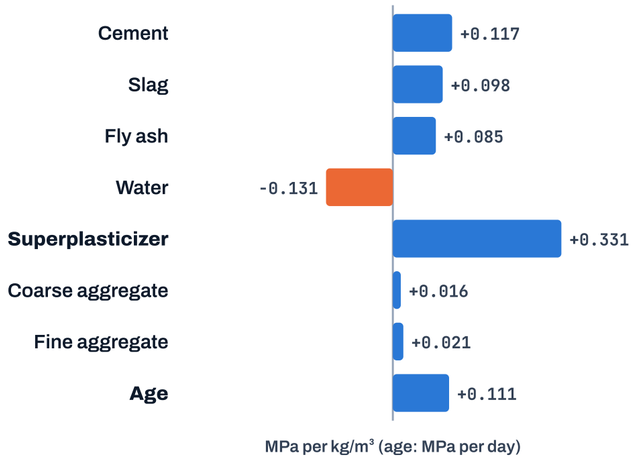


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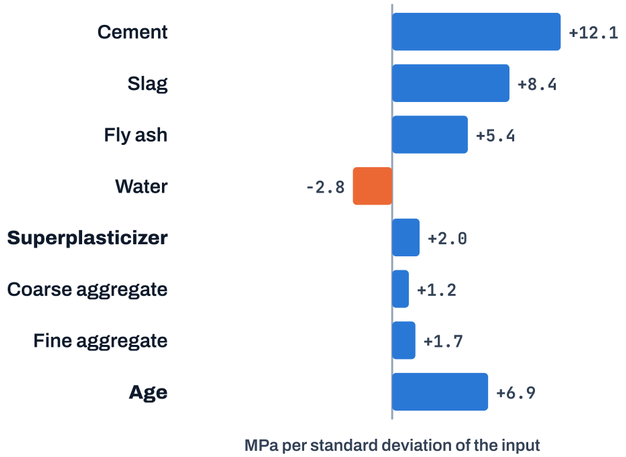
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Same model, standardised inputs

Change per one standard deviation of each input.

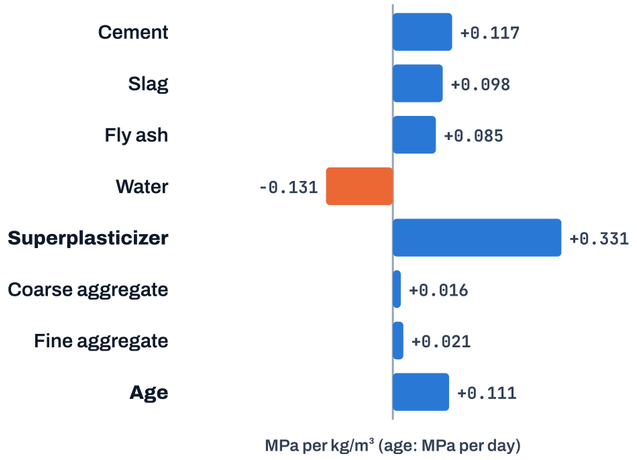


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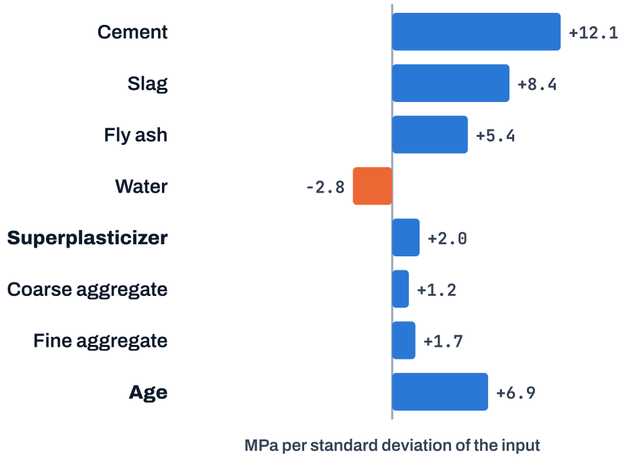
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READING ONE

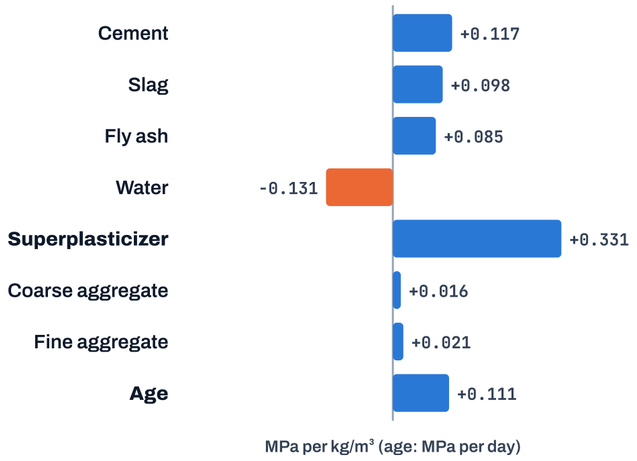
Age +0.111: within this model, one extra day adds about 0.111 MPa, holding the other inputs constant.

LOOKING INSIDE THE MODEL

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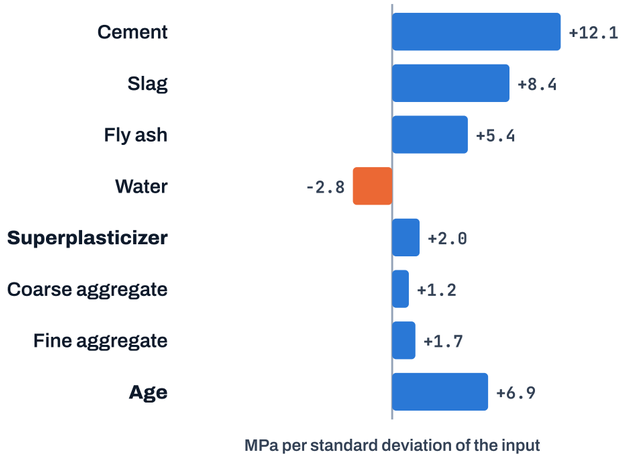
Raw coefficients

Change in predicted strength per unit of each input, others held constant.



Same model, standardised inputs

Change per one standard deviation of each input.



READING ONE

Age +0.111: within this model, one extra day adds about 0.111 MPa, holding the other inputs constant.

BUT BE CAREFUL

Units matter: superplasticizer has the largest raw weight but one of the smallest effects per standard deviation.

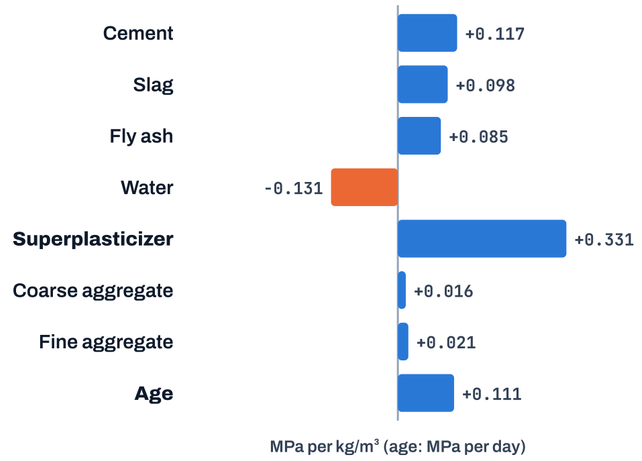
A larger coefficient does not automatically mean greater importance. A coefficient shows **association in the fitted model**, not causation.

LOOKING INSIDE THE MODEL

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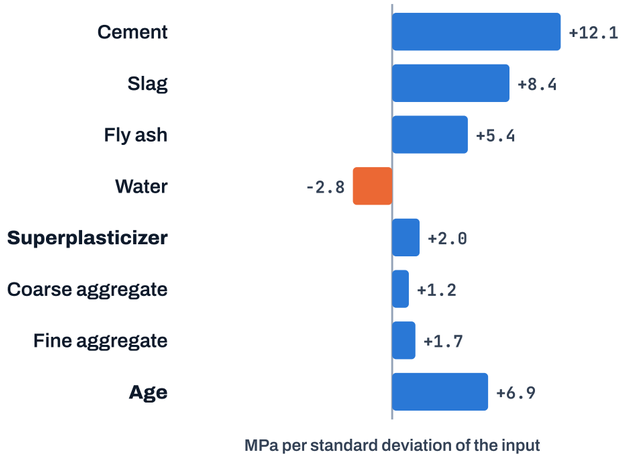
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This is an important engineering judgement point. More in Session 07.

SOME MATHS BEHIND ML MODELS · 1 OF 5

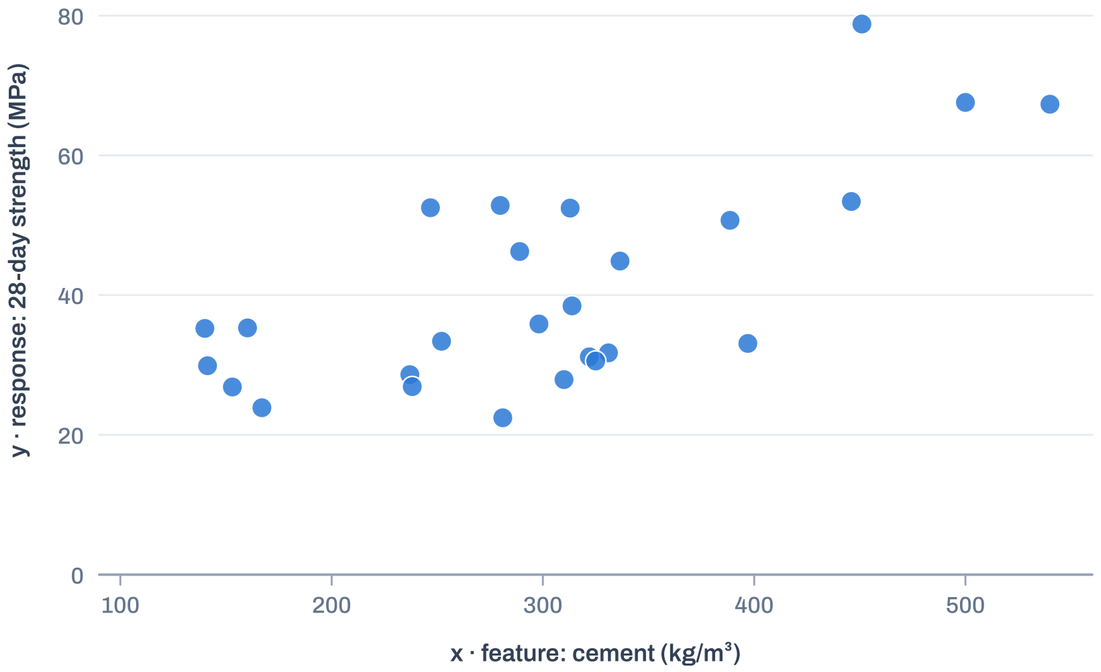
Linear regression: the base model of statistical ML

THE MODEL · ONE INPUT

$$f_{w,b}(x^{(i)}) = w\,x^{(i)} + b$$

Parameters: slope w and intercept b.

26 real 28-day mixes, one input: cement



Notation after DeepLearning.AI (Machine Learning Specialization)

SOME MATHS BEHIND ML MODELS · 1 OF 5

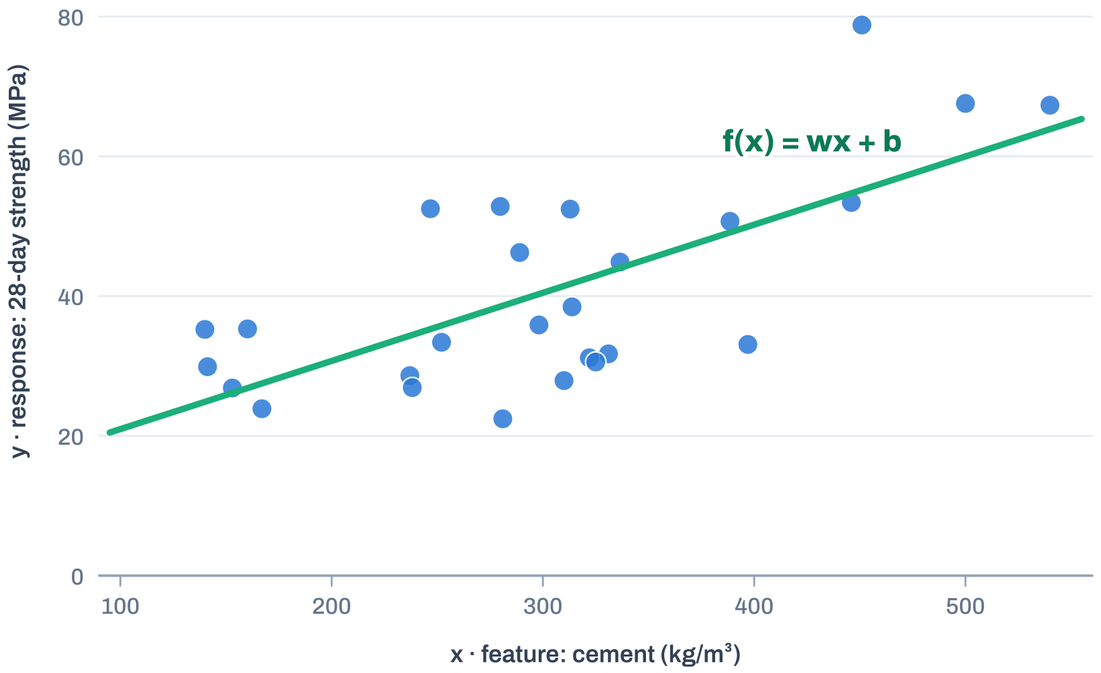
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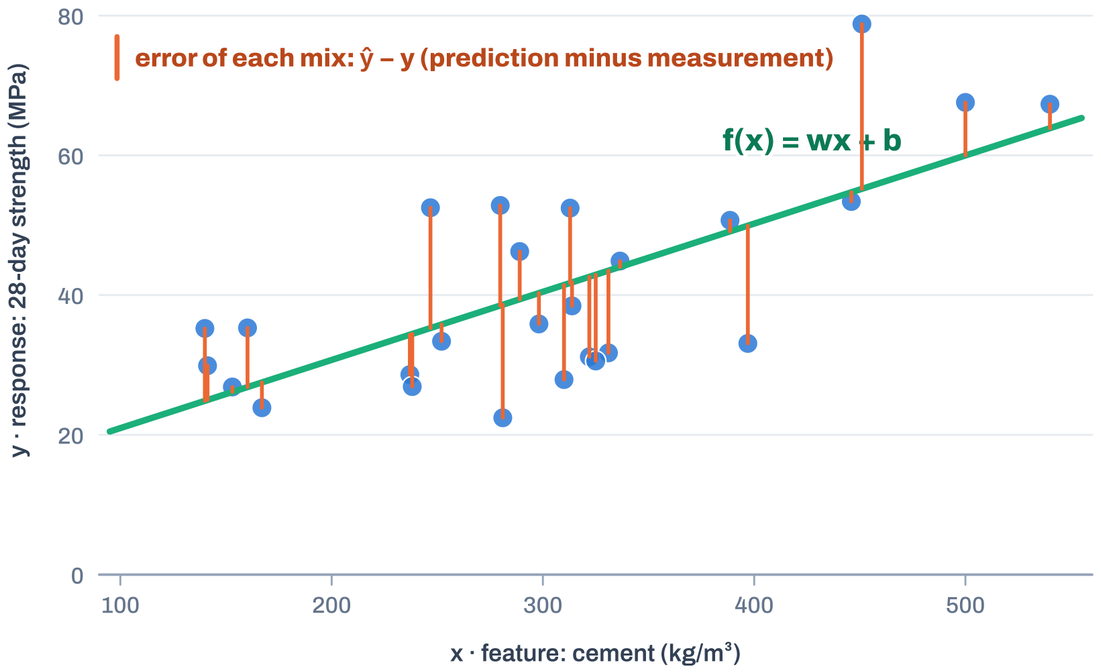
Parameters: slope w and intercept b.

THE COST · SQUARED-ERROR COST FUNCTION

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)} \right)^2$$

Square every error, average over the m examples.

26 real 28-day mixes, one input: cement



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SOME MATHS BEHIND ML MODELS · 1 OF 5

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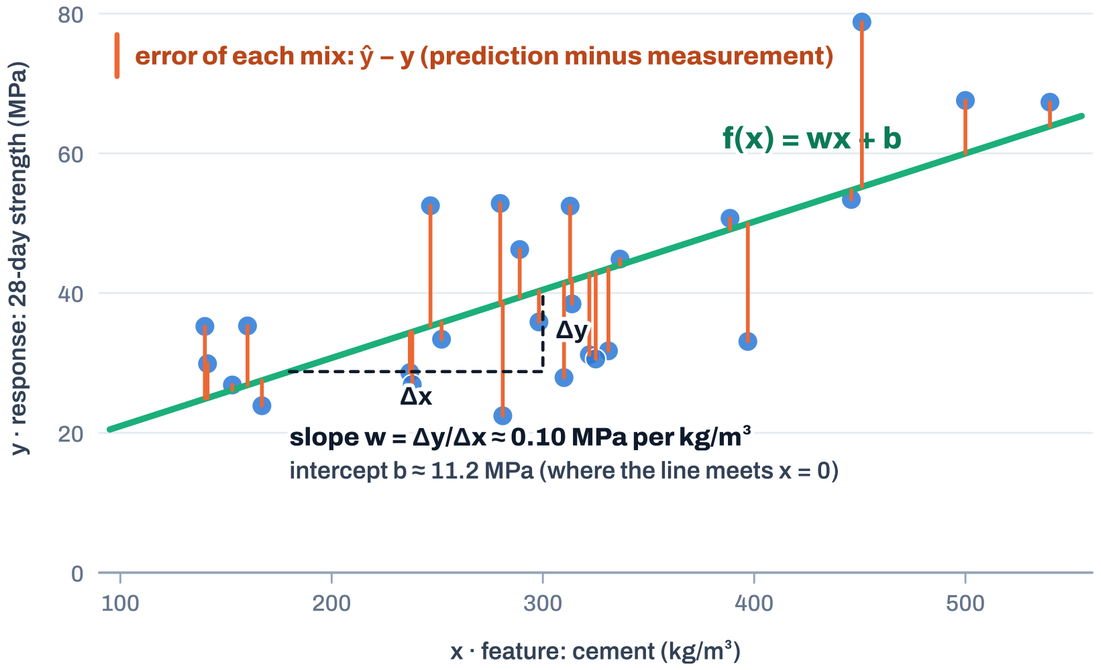
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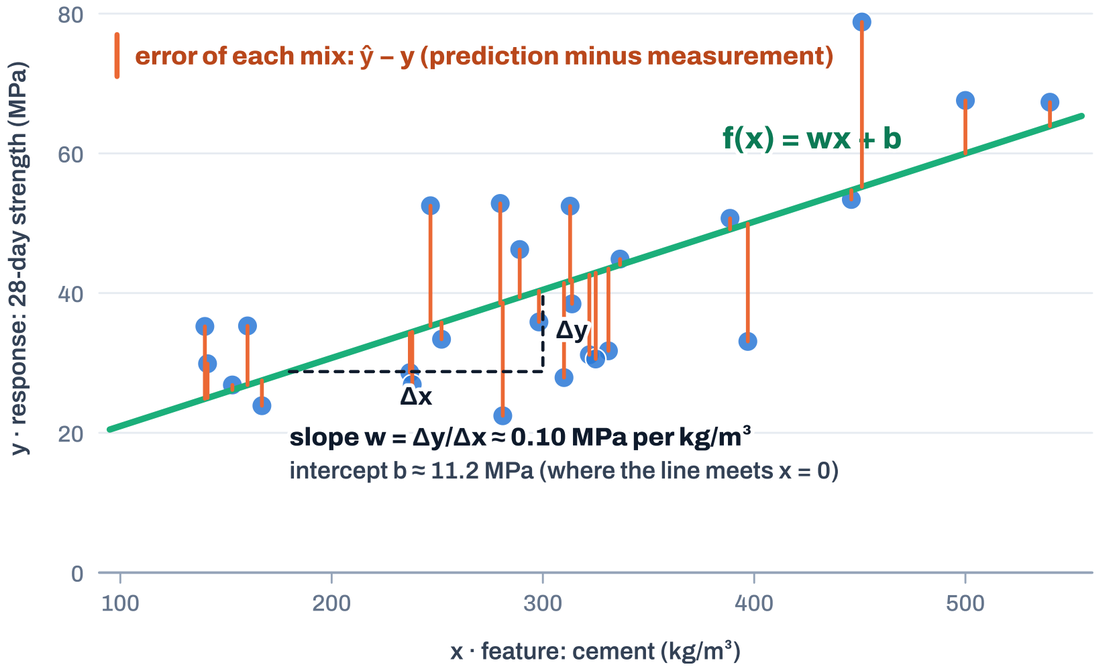
Square every error, average over the m examples.

THE GOAL

minimise $J(w, b)$ over w, b

With 8 inputs it is the same idea: one w per input.

26 real 28-day mixes, one input: cement



Notation after DeepLearning.AI (Machine Learning Specialization)

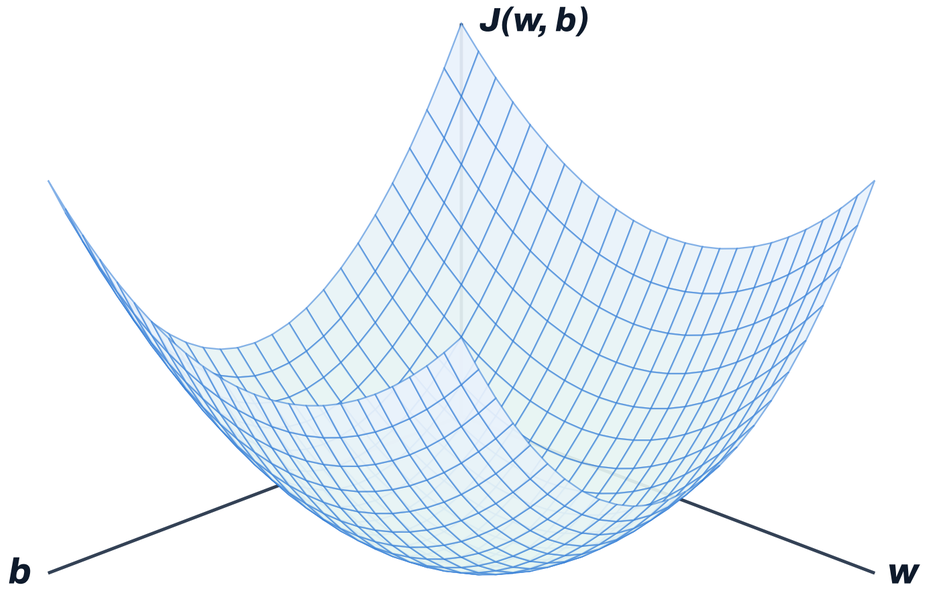
SOME MATHS BEHIND ML MODELS · 2 OF 5

The cost function is a bowl. We walk downhill.

CONVEX

For linear regression, $J(w, b)$ has a single lowest point: no false valleys.

GRADIENT DESCENT, IN OUTLINE



Cost surface computed from the 26 mixes of the previous slide (cement standardised). Concept after DeepLearning.AI.

SOME MATHS BEHIND ML MODELS · 2 OF 5

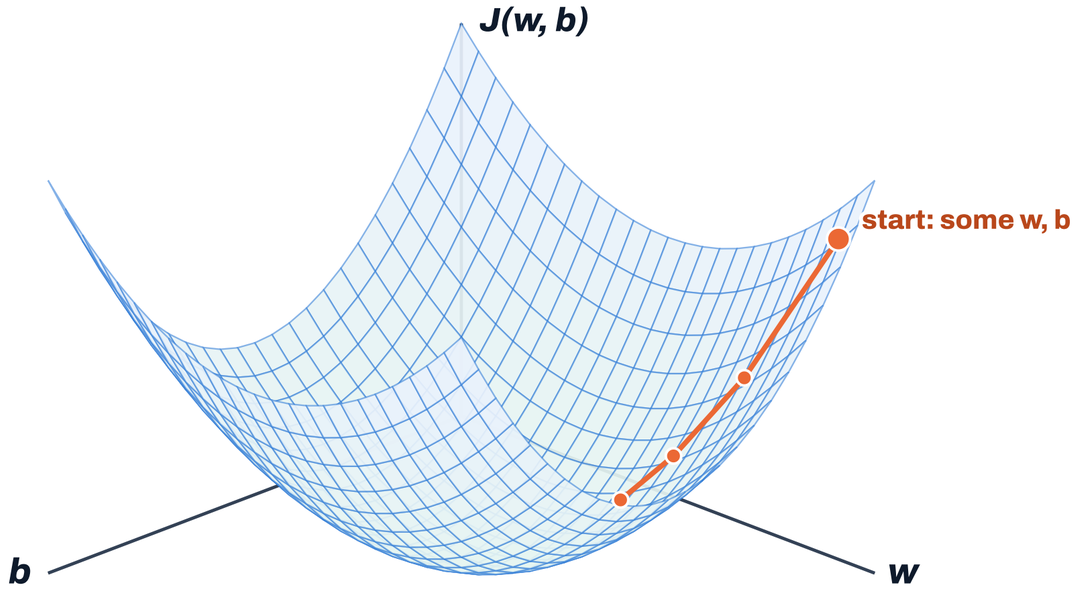
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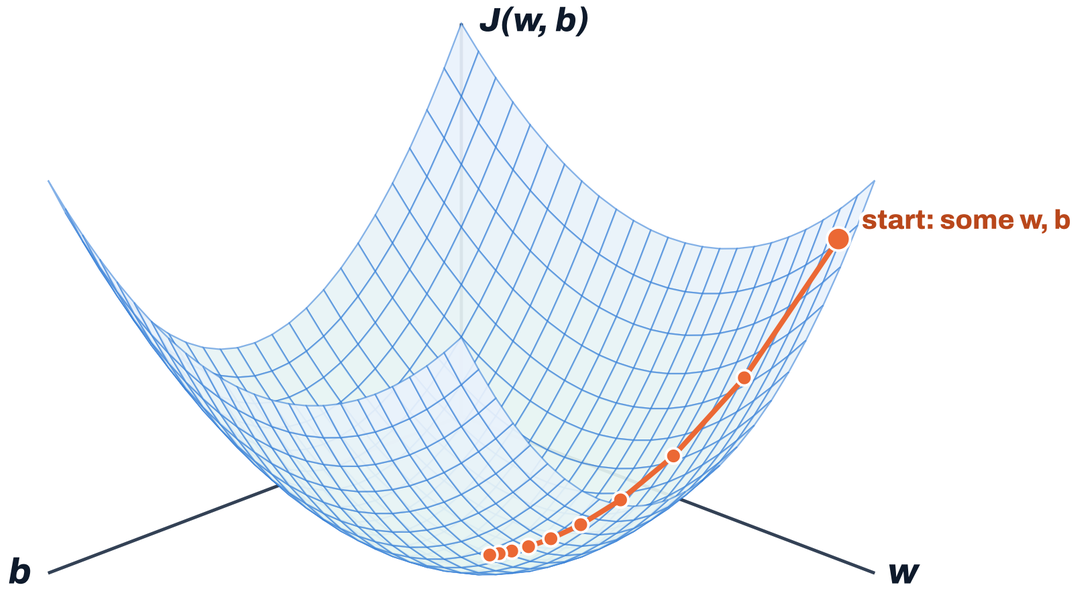
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GRADIENT DESCENT, IN OUTLINE

01 Start with some w, b

02 Keep changing w, b to reduce $J(w, b)$



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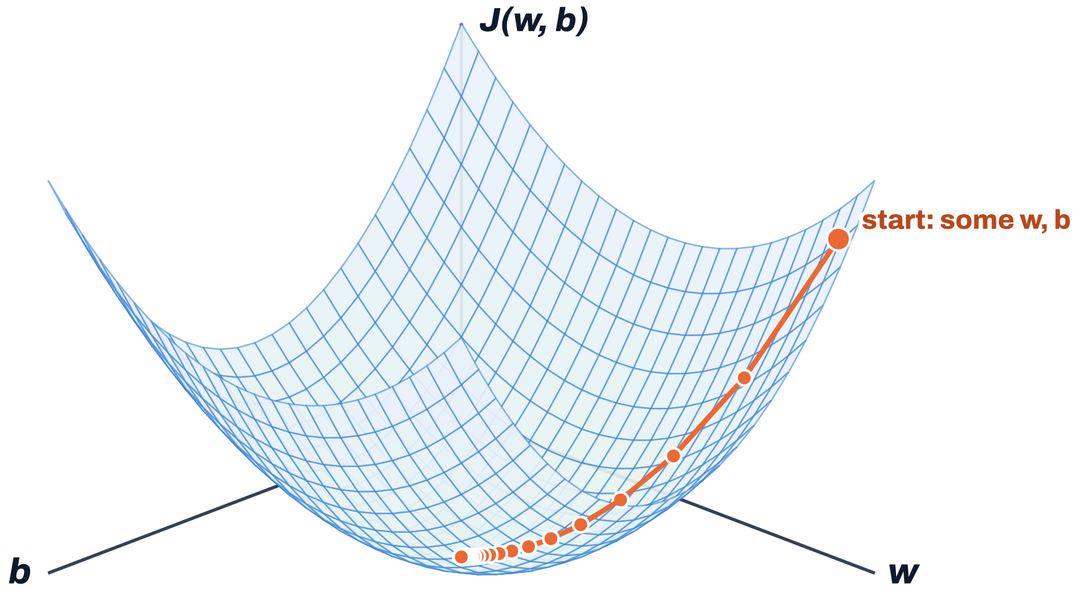
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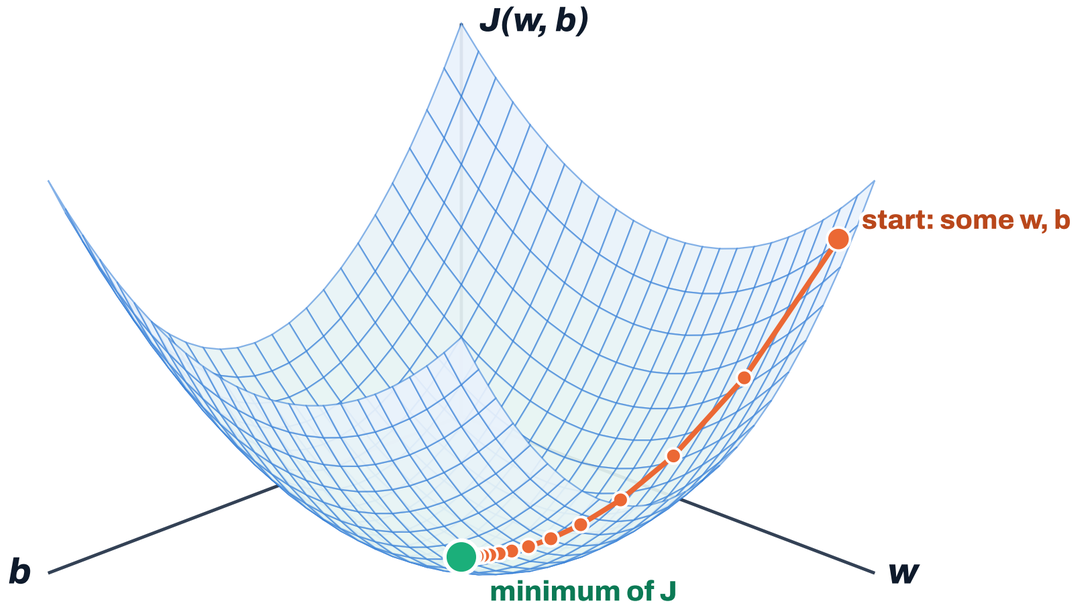
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01 Start with some w, b

02 Keep changing w, b to reduce $J(w, b)$

03 Until we settle at or near a minimum



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SOME MATHS BEHIND ML MODELS · 2 OF 5

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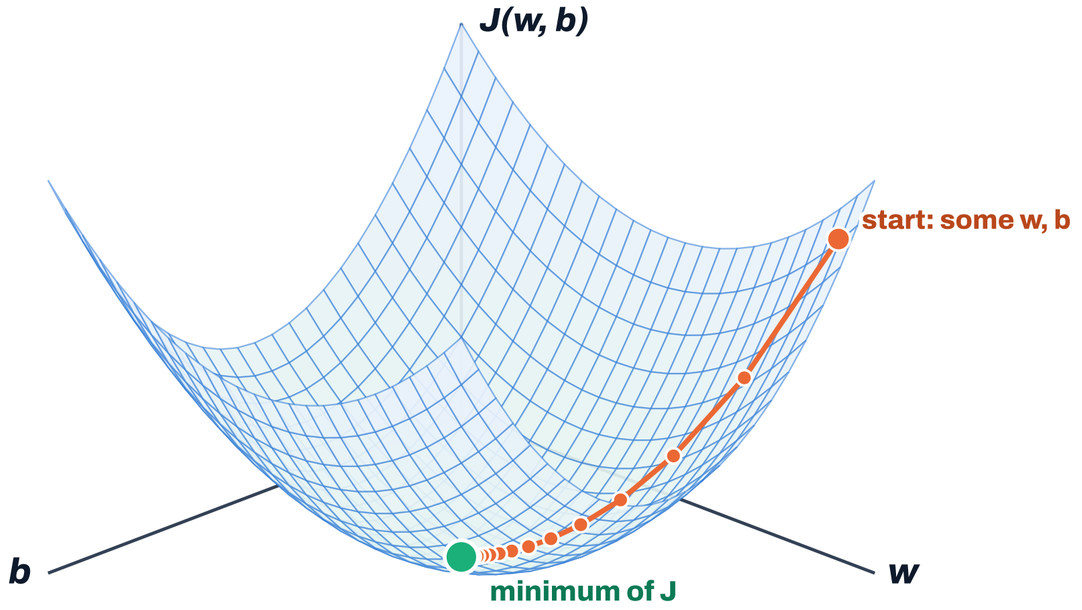
GRADIENT DESCENT, IN OUTLINE

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Here: 40 steps take the cost from **1,103** to **54** MPa^2.



Cost surface computed from the 26 mixes of the previous slide (cement standardised). Concept after DeepLearning.AI.

SOME MATHS BEHIND ML MODELS · 3 OF 5

Gradient descent: follow the slope downhill

Repeat until convergence:

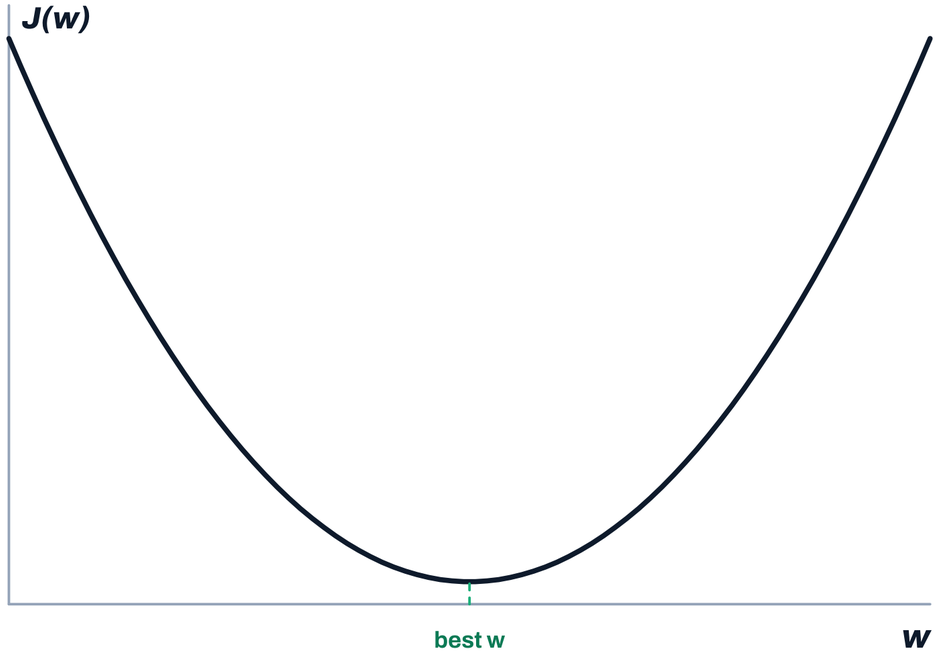
$$w = w - \alpha \frac{\partial}{\partial w} J(w,b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w,b)$$

α learning rate: step size

derivative: slope of J

$J(w)$ with b at its best value



Notation after DeepLearning.AI; concept also in CS771, IIT Kanpur

SOME MATHS BEHIND ML MODELS · 3 OF 5

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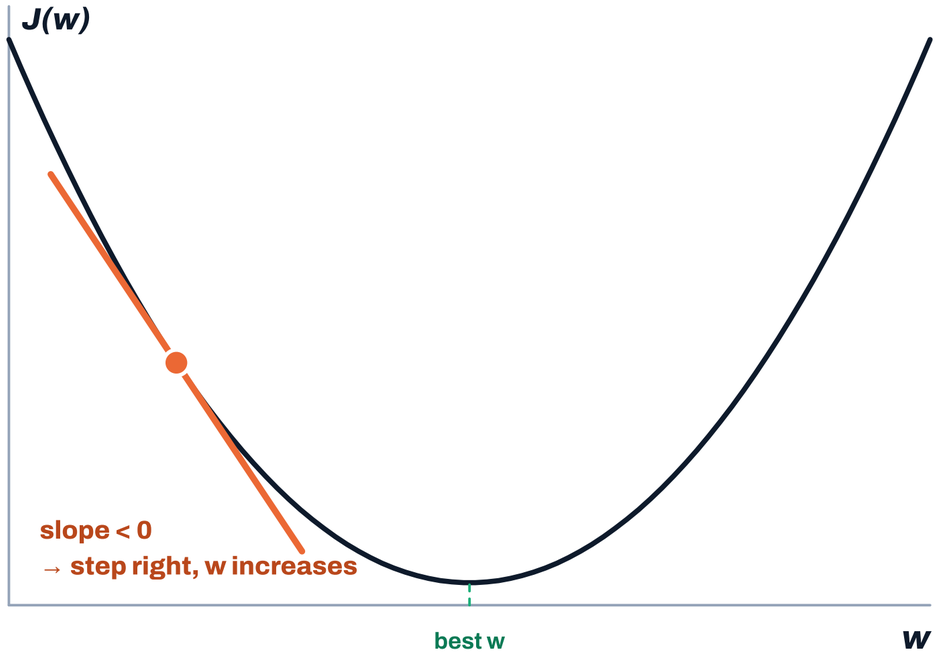
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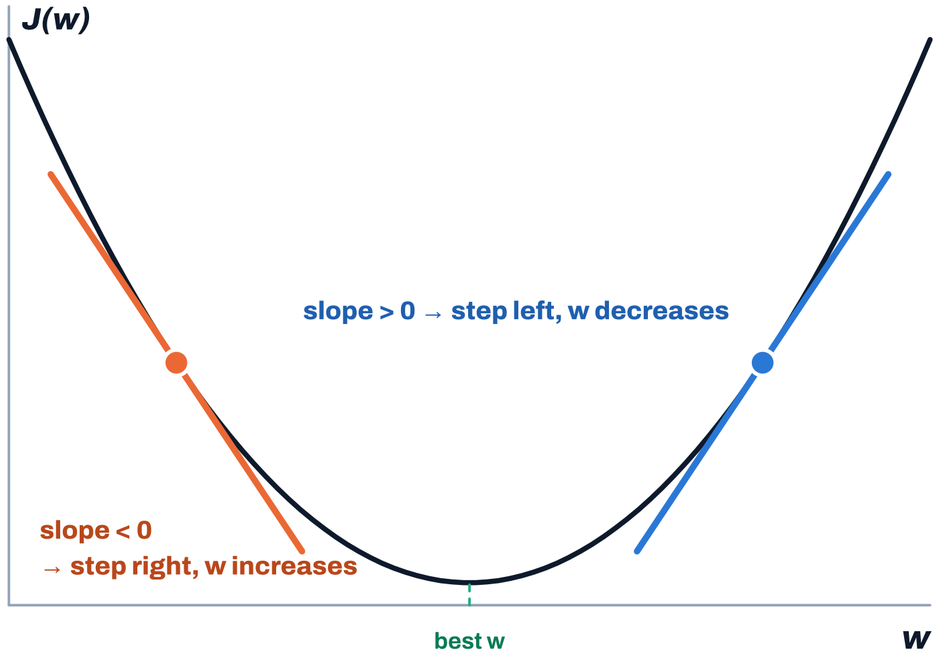
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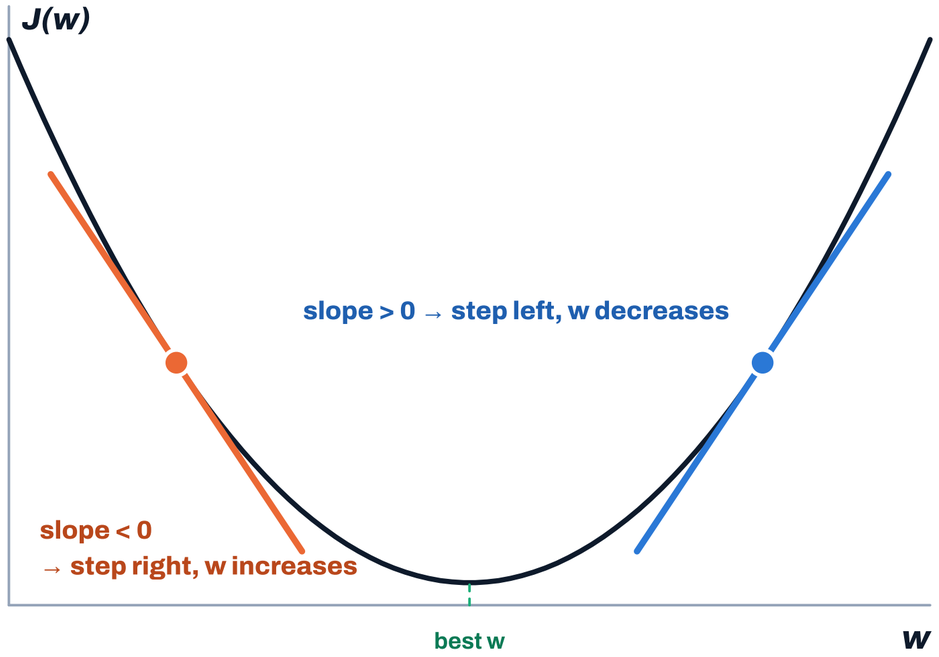
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WHY IT WORKS

A negative slope makes *w* grow; a positive slope makes it shrink. Steps get smaller as the slope flattens near the minimum.

$J(w)$ with b at its best value



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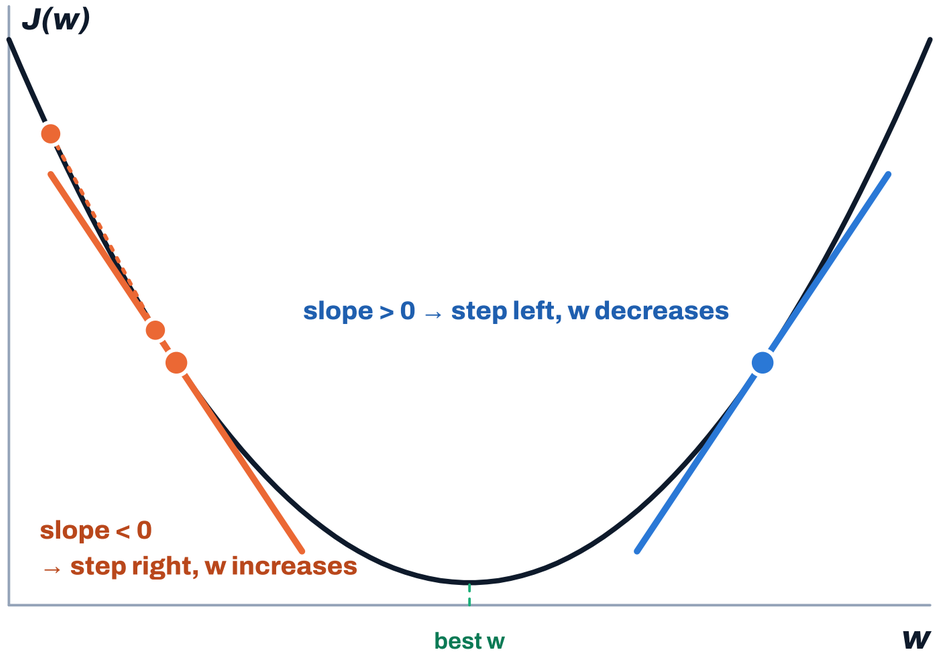
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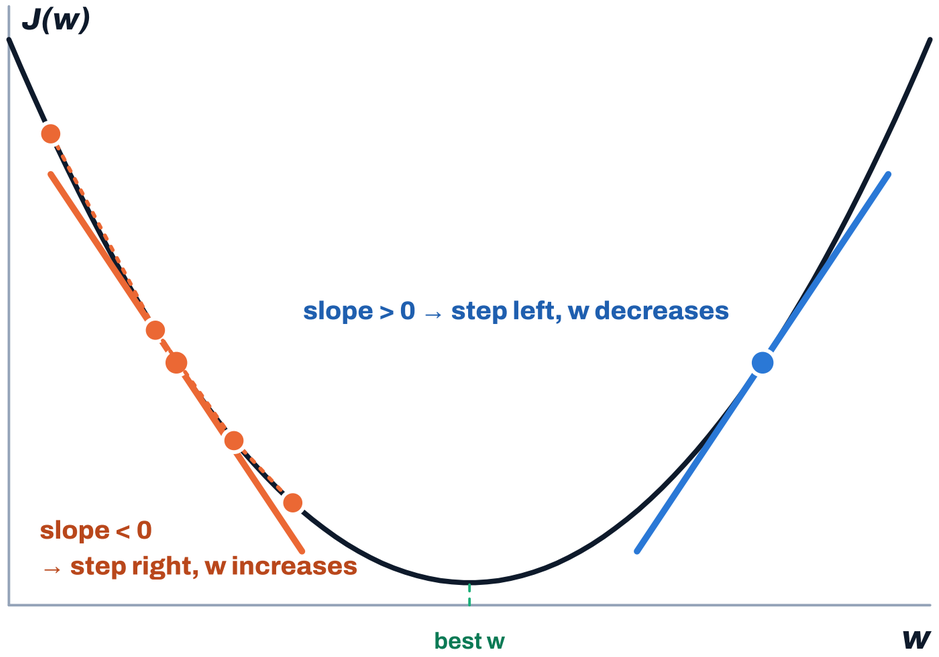
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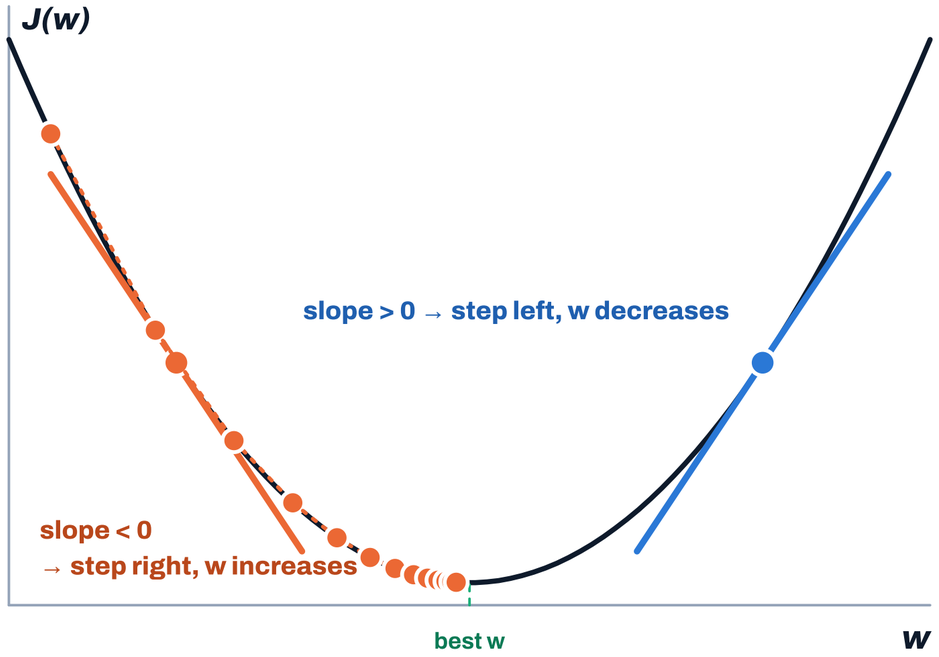
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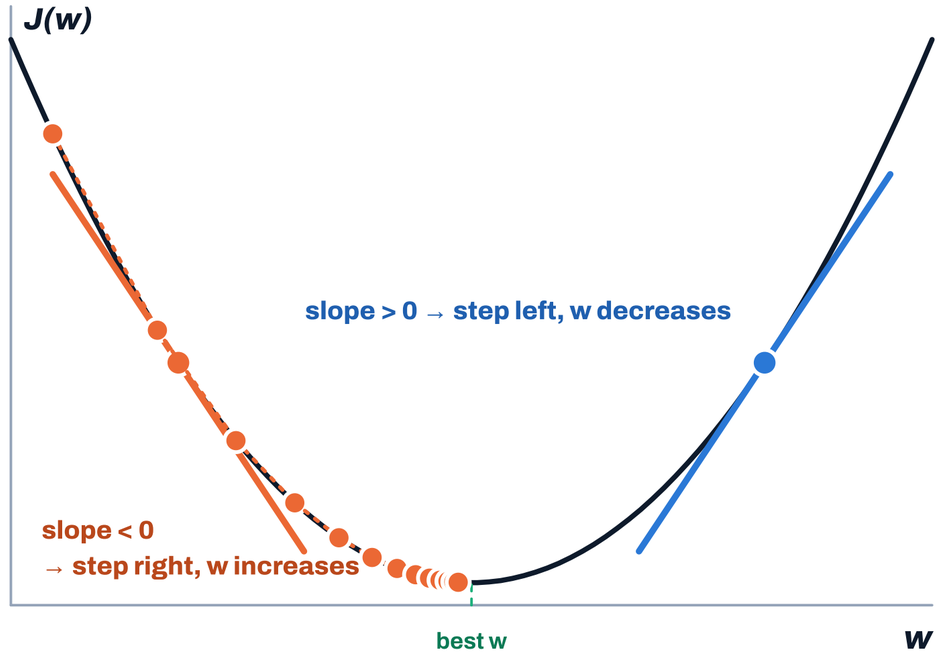
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WHY IT WORKS

A negative slope makes *w* grow; a positive slope makes it shrink. Steps get smaller as the slope flattens near the minimum.

Update *w* and *b* **together**, using the slope at the current point.

J(*w*) with *b* at its best value



Notation after DeepLearning.AI; concept also in CS771, IIT Kanpur

SOME MATHS BEHIND ML MODELS · 4 OF 5 · HYPERPARAMETER TUNING

The learning rate α decides whether we arrive

All three runs start from the same point on the same cost curve as the previous slide.

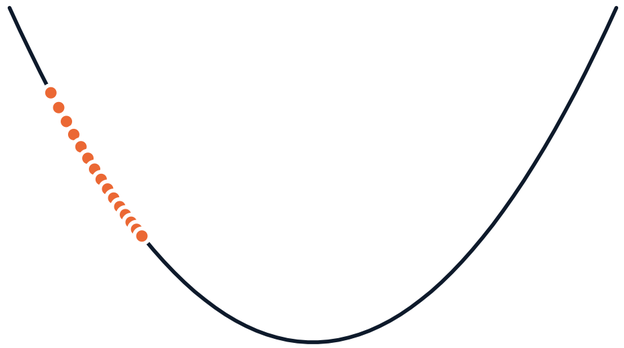
SOME MATHS BEHIND ML MODELS · 4 OF 5 · HYPERPARAMETER TUNING

The learning rate α decides whether we arrive

α = 0.03 · TOO SMALL

Slow

Tiny steps: after 14 steps we have covered only 35% of the way.



Converges, eventually

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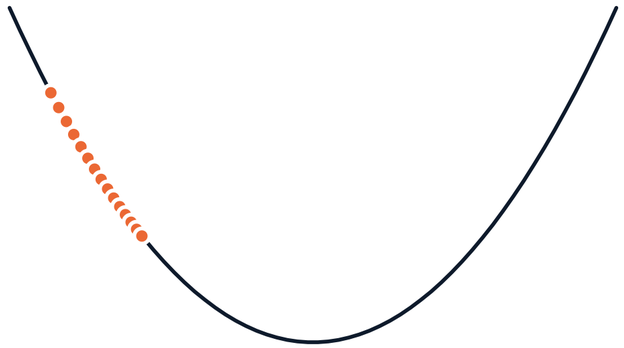
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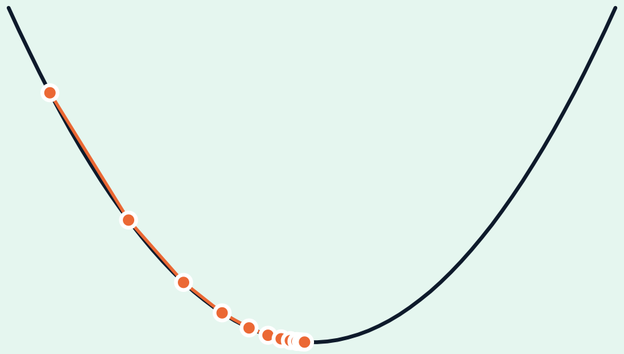


Converges, eventually

α = 0.3 · ABOUT RIGHT

Efficient

Large steps far away, small steps near the minimum: there in about 10 steps.



Converges quickly

All three runs start from the same point on the same cost curve as the previous slide.

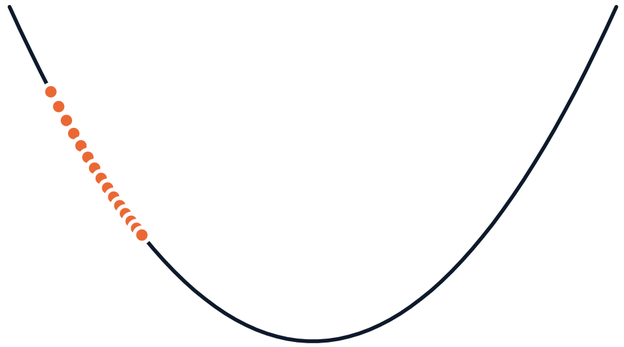
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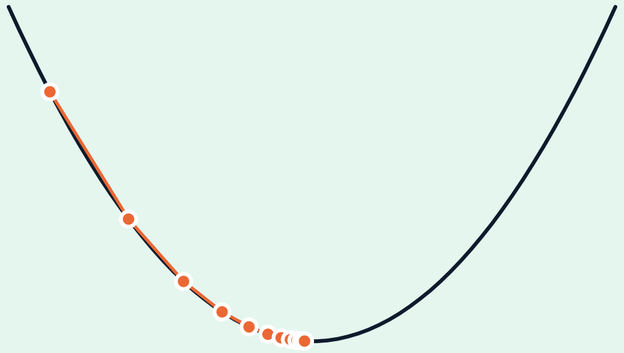


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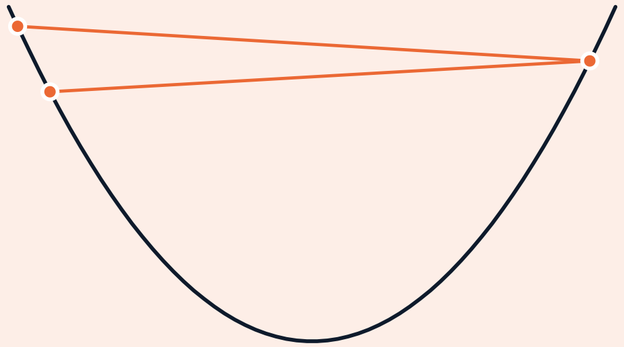


Converges quickly

α = 2.06 · TOO LARGE

Overshoots

Each step jumps past the minimum and lands higher up the other side.



Fails to converge

All three runs start from the same point on the same cost curve as the previous slide.

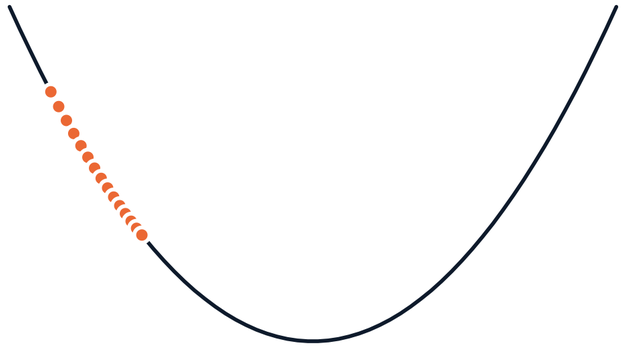
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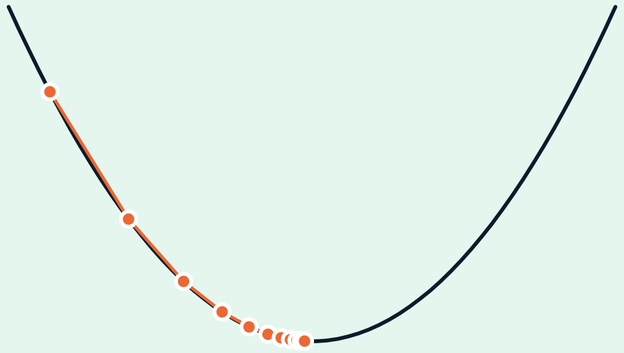


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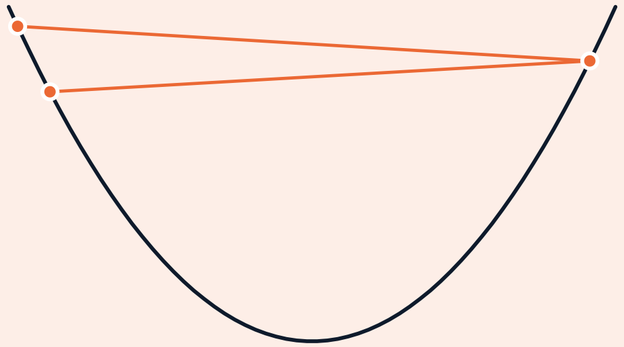


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α = 2.06 · TOO LARGE

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Each step jumps past the minimum and lands higher up the other side.



Fails to converge

α is a hyperparameter: we choose it; the model does not learn it. scikit-learn's LinearRegression solves for the weights directly, so today we do not tune α.

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SOME MATHS BEHIND ML MODELS · 5 OF 5 · DATA PRE-PROCESSING

Feature scaling: make the bowl round

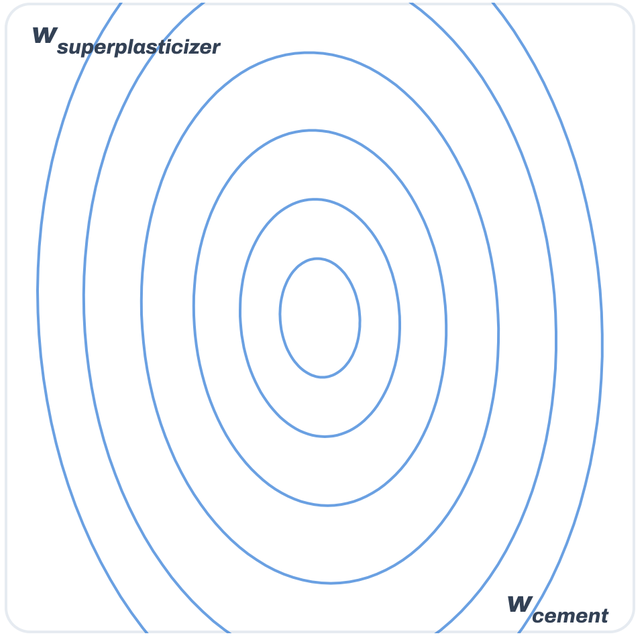
Once the features are selected, they may not be on the same scale: cement varies by about 104 kg/m³, superplasticizer by about 6.

PROBLEMATIC

Features with larger scales can dominate comparisons, and gradient descent crawls along a long, narrow valley.

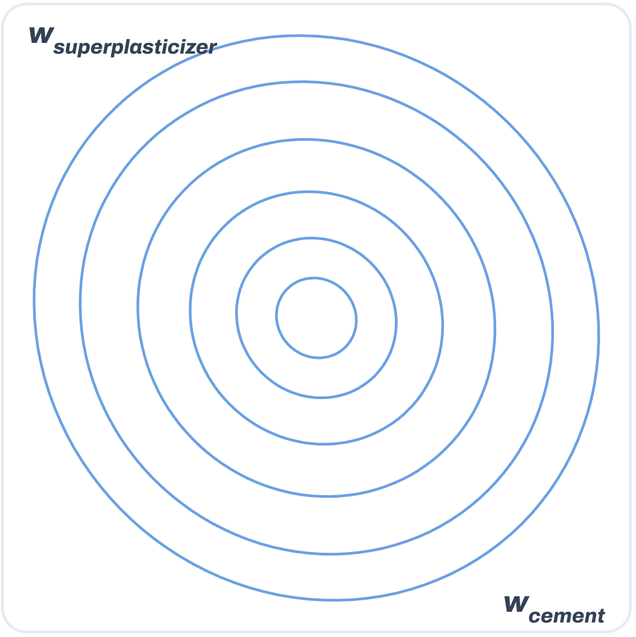
GRADIENT DESCENT WITHOUT SCALING

Raw kg/m³



GRADIENT DESCENT AFTER SCALING

Standardised



Real cost contours for predicting strength from cement and superplasticizer (1,005 mixes), same safe learning rate rule, stopping at 0.1% of the starting cost. Idea after Virginia Tech ECE 5424 (Lec. 03) and DeepLearning.AI.

SOME MATHS BEHIND ML MODELS · 5 OF 5 · DATA PRE-PROCESSING

Feature scaling: make the bowl round

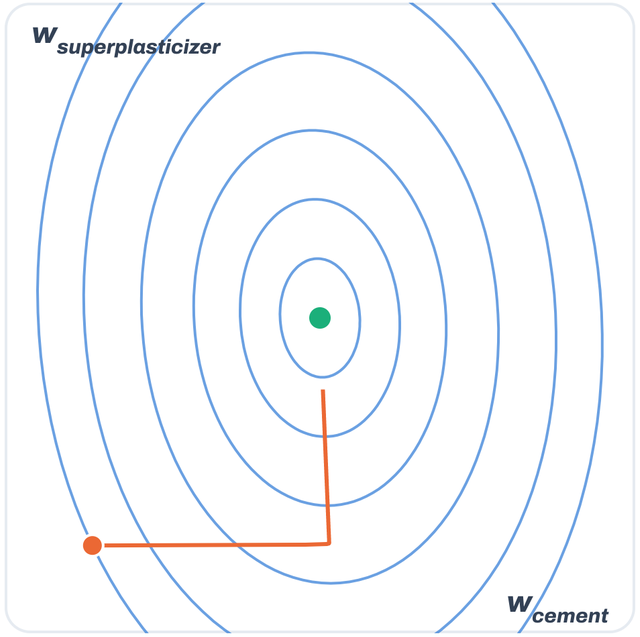
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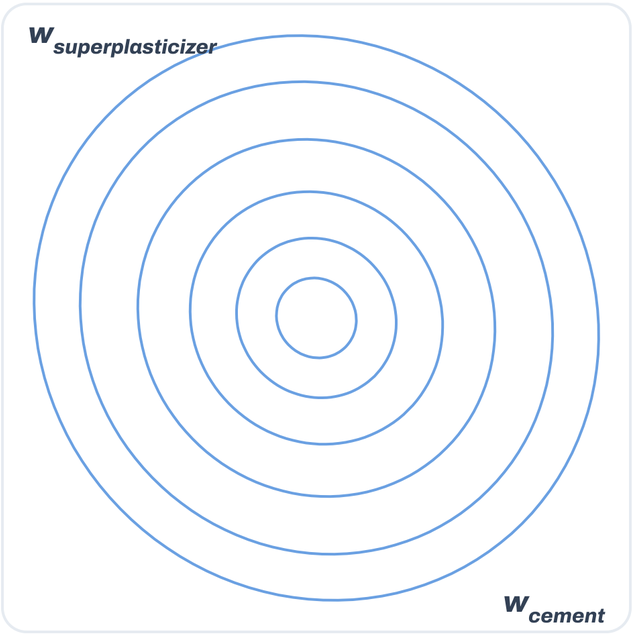
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984 iterations

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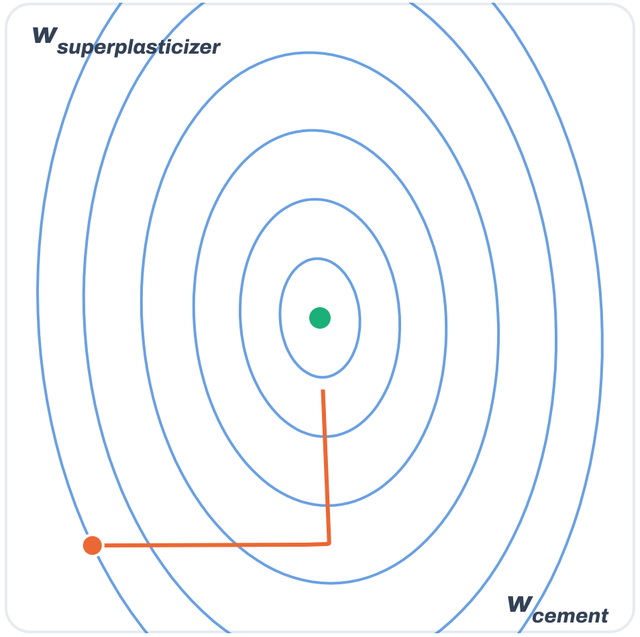
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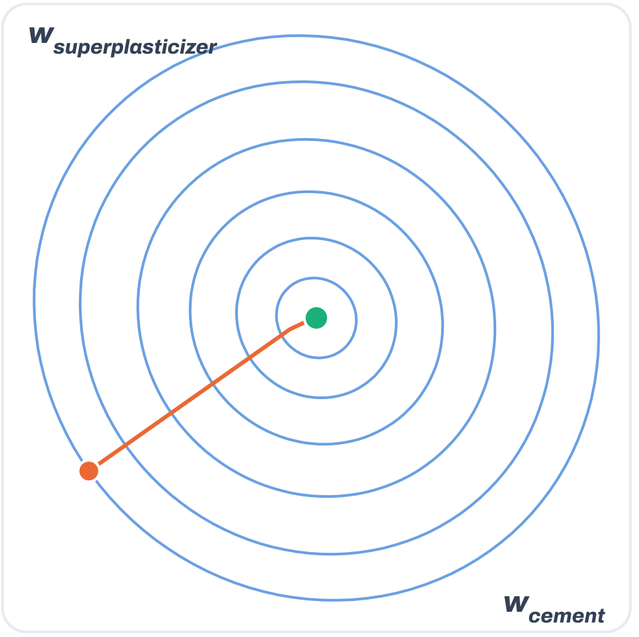
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GRADIENT DESCENT AFTER SCALING

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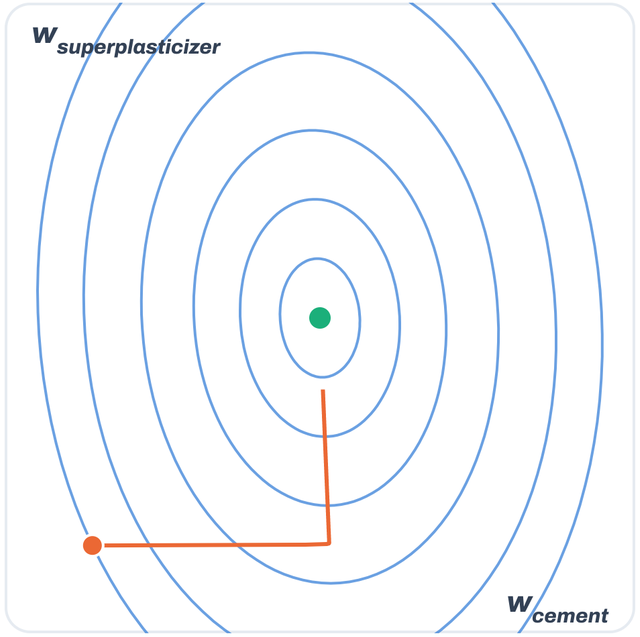
THEREFORE, STANDARDISE

Mean 0, standard deviation 1 for every feature. It stabilises the optimisation used in ML algorithms.

```
StandardScaler().fit(X_train)
```

GRADIENT DESCENT WITHOUT SCALING

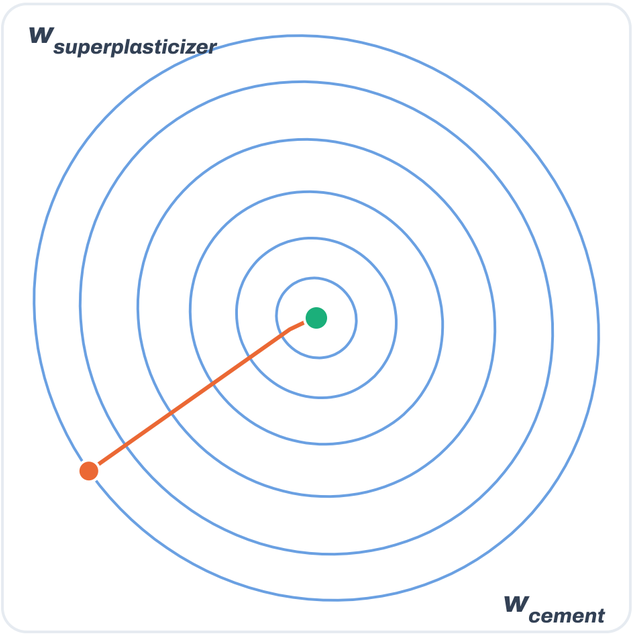
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02

a A scorecard: MAE, RMSE, R^2

b Which metric should we trust?

c Train vs test performance

d Parity plot

e Residuals: where is it wrong?

PART 02 · EVALUATION

How good is our model?

PREDICTION IS NOT ENOUGH

A model can make predictions. How good are they?

We need a scorecard. Today: three regression metrics.

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Coefficient of Determination

no unit · higher is better, max 1

How much better than predicting the mean?

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Different metrics answer slightly different questions.

METRIC 1 · ON OUR 201 TEST MIXES

MAE: Mean Absolute Error

QUESTION · ON AVERAGE, HOW LARGE IS OUR PREDICTION ERROR?

$$\text{MAE} = \frac{1}{m}\sum_{i=1}^{m} | y_i - \hat{y}_i |$$

actual target value · predicted target

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LINEAR REGRESSION · TEST SET

8.9 MPa

“Our predictions are off by about 8.9 MPa on average.”

Mean baseline: **14.0 MPa**. The model cuts the average error by 36%.

METRIC 2 · LARGE ERRORS ARE PENALISED MORE STRONGLY

RMSE: Root Mean Squared Error

SQUARES THE ERRORS BEFORE AVERAGING

$$\text{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}\left(y_i - \hat{y}_i\right)^2}$$

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WHY SQUARING MATTERS · ERRORS OF TWO MODELS ON 4 MIXES

A	3	3	3	3	→ MAE 3.0 · RMSE 3.0
B	0	0	0	12	→ MAE 3.0 · RMSE 6.0

MAE → average error size

RMSE → average error with a greater penalty for large misses

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LINEAR REGRESSION · TEST SET

11.2 MPa

RMSE ≥ MAE always. Here the gap (11.2 vs 8.9) says some mixes are missed by a lot.

Mean baseline: **17.3 MPa**. Lower is better.

METRIC 3 · HOW MUCH VARIATION DOES THE MODEL EXPLAIN?

R^2: compared with the mean baseline

MODEL ERROR RELATIVE TO BASELINE ERROR

$$R^2 = 1 - \frac{\Sigma (y_i - \hat{y}_i)^2}{\Sigma (y_i - \bar{y})^2}$$

$\bar{y}$ arithmetic mean of y = the baseline

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perfect predictions

$R^2 = 0$
no better than the mean

$R^2 < 0$
worse than the mean, possible on unseen data

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$R^2 = 0.58$ does not mean “58% of predictions are correct”. It means the squared error is 58% smaller than the mean baseline’s.

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LINEAR REGRESSION · TEST SET

0.58

The model explains well over half the variation in strength that the mean leaves unexplained.

Mean baseline: **0.00**, by construction.

DON’T ASK “WHICH METRIC IS BEST?” · ASK WHAT EACH ONE TELLS US

Which metric should we trust?

METRIC	WHAT IT TELLS US	MEAN BASELINE	LINEAR REGRESSION

All values on the 201 test mixes (random_state = 42).

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ENGINEERING JUDGEMENT

A good R^2 can still hide errors too large for the application. Is ±8.9 MPa acceptable for your design decision?

A metric is evidence, not judgement.

All values on the 201 test mixes (random_state = 42).

TRAINING PERFORMANCE ≠ GENERALISATION

Does the model work on new data?

EVALUATED ON	MAE	RMSE	R^2

MAE and RMSE in MPa. Training: 804 mixes; test: 201 mixes.

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→ encouraging

Much better on training than test
→ possible overfitting / poor generalisation

Poor or modest on both
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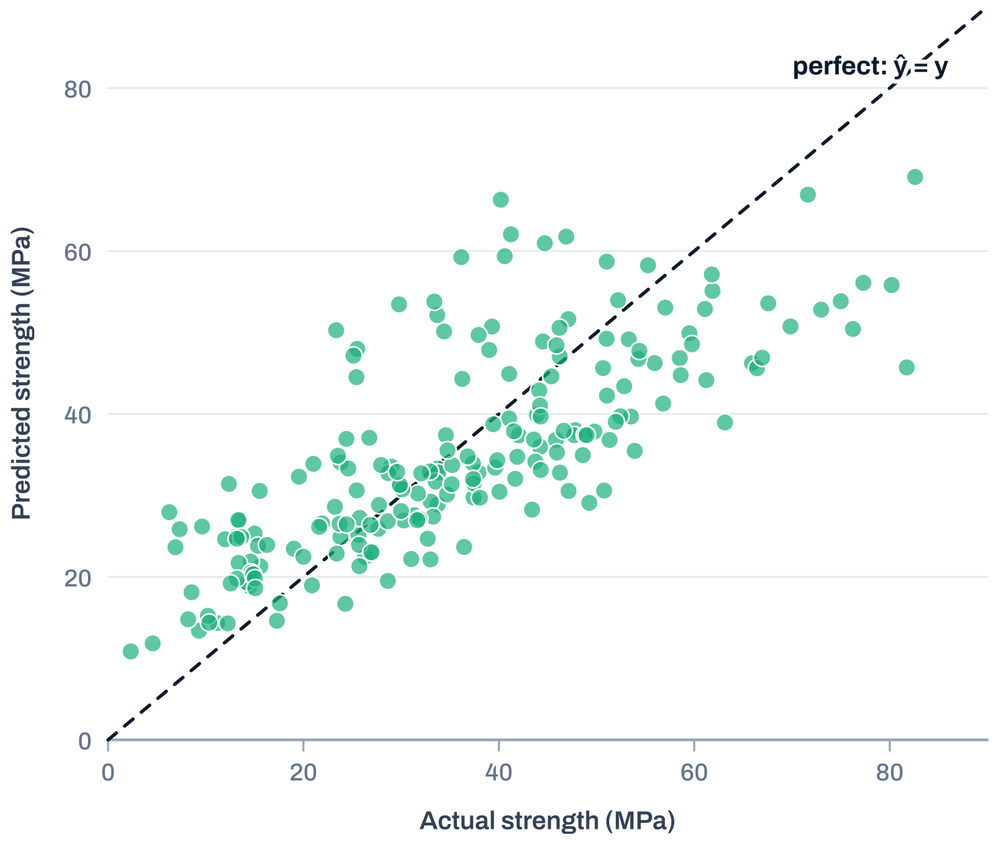
⚠ **Good training performance alone is not enough.**

MAE and RMSE in MPa. Training: 804 mixes; test: 201 mixes.

VISUALISING PREDICTIONS

Parity plot: actual vs predicted

201 test mixes



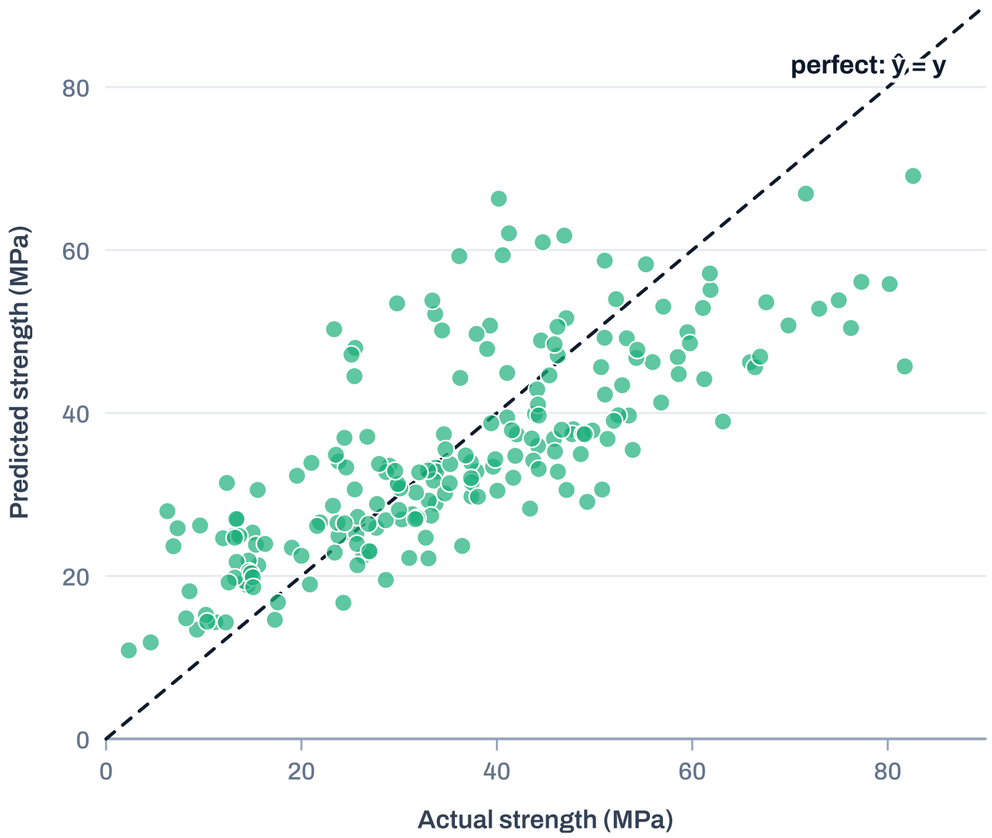
IF PREDICTIONS WERE PERFECT

Every point would lie on the 45° line.

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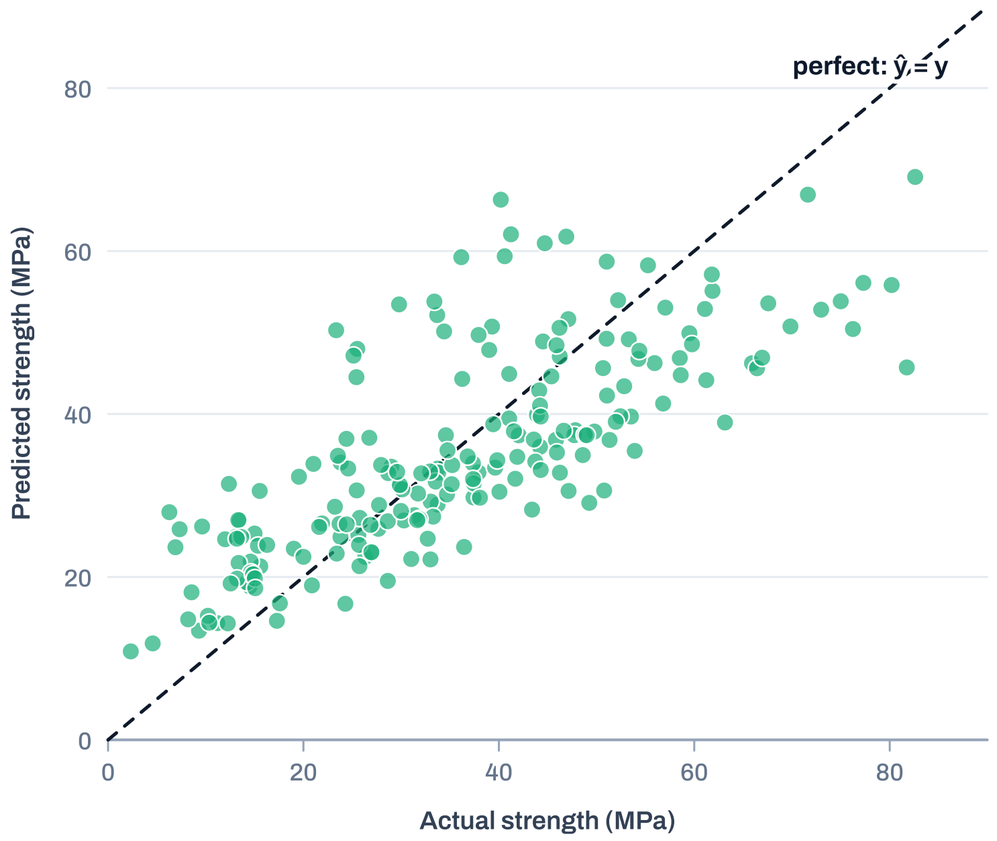
LOOK FOR

- How close points are to the line
- Systematic deviations
- Extreme errors
- Different behaviour in different ranges

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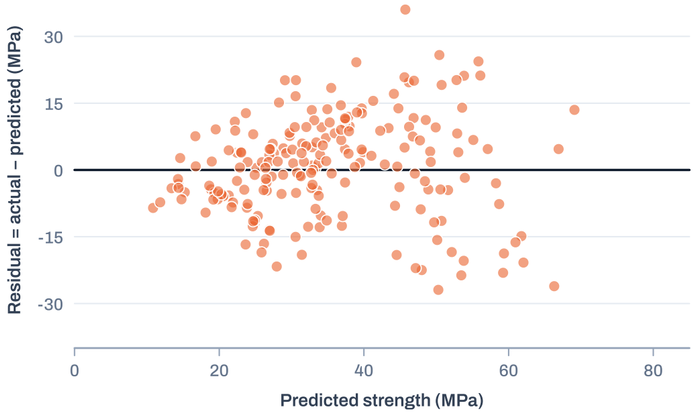
Strong mixes above 60 MPa sit below the line: the model **under-predicts high strength.**

RESIDUAL = ACTUAL – PREDICTED · WHERE AND WHY IS THE MODEL WRONG?

Residuals help us diagnose the model

Against the prediction

Ideally: centred on zero, no pattern.



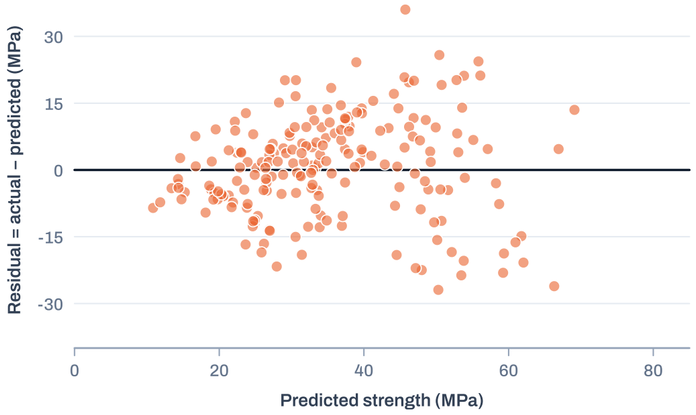
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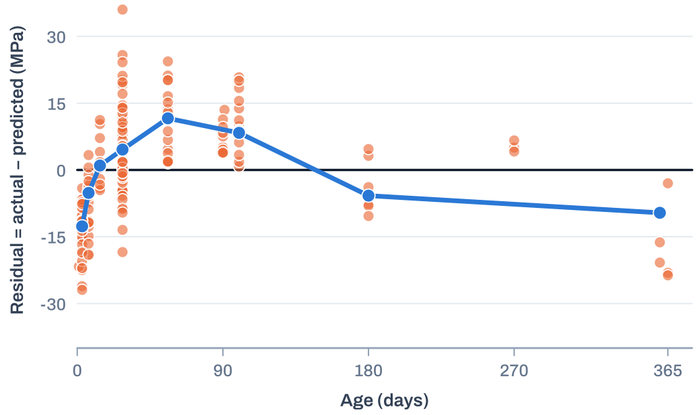
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Against age: a clear curve

Blue: median residual at each test age.



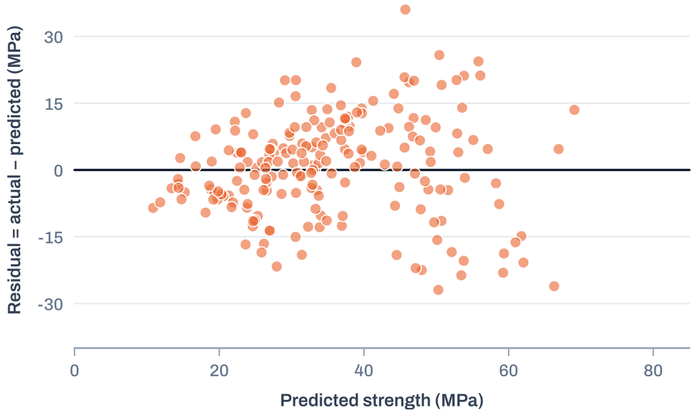
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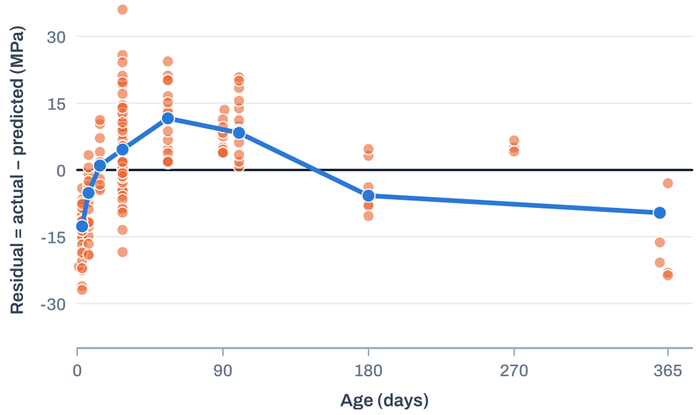
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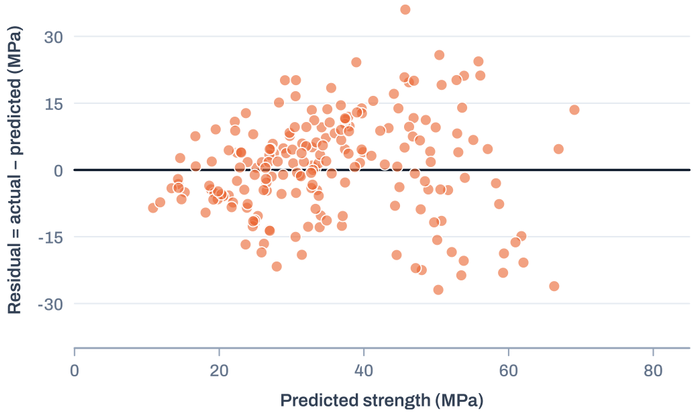
→ the linear model misses nonlinear behaviour. Here: –12.7 MPa at 3 days, +11.6 at 56, –9.6 at a year.

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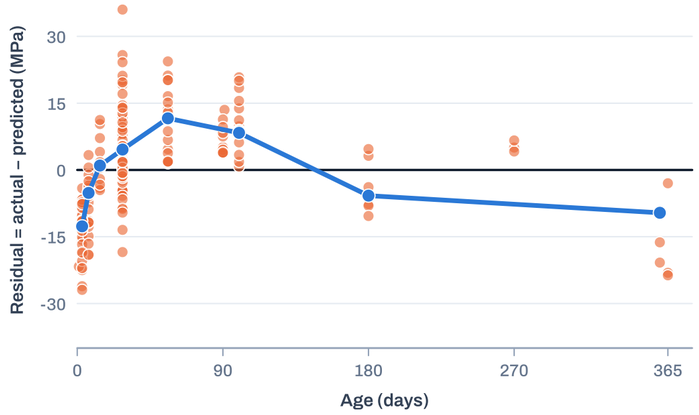
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→ the error changes with the target or input range

Large isolated residuals

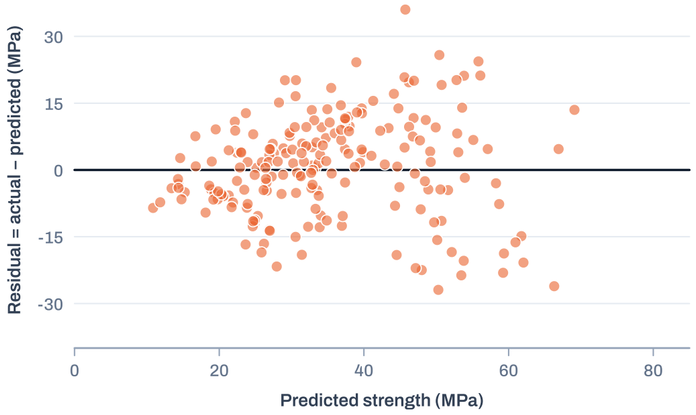
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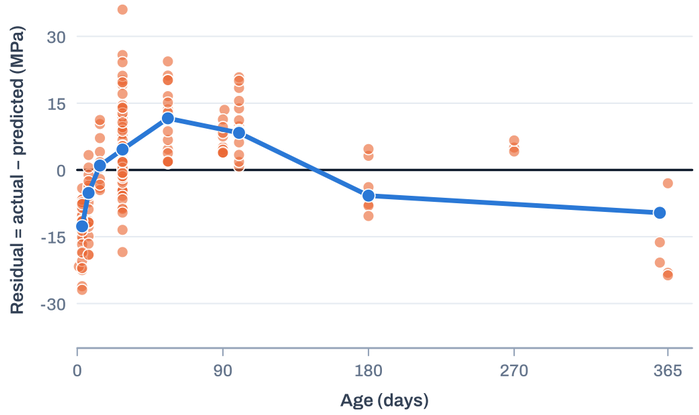
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Don’t just ask “how good is the score?” Ask “where and why is the model wrong?”

03

PART 03 · VALIDATION & JUDGEMENT

Is it consistent, and can we trust it?

a One split is only one split

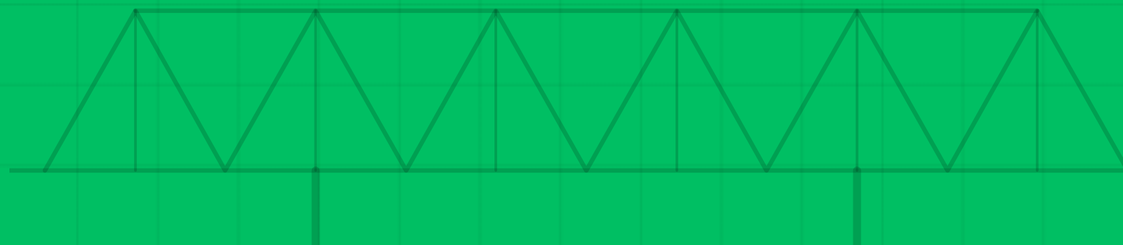
b K-fold cross-validation

c Look beyond the mean

d The test-set rule

e From performance to engineering judgement

f Your turn: diagnose the model



ONE SPLIT IS ONLY ONE SPLIT

What if we change the split? Same model, same data.

```
random_state = 42
```

R^2 0.58

ONE SPLIT IS ONLY ONE SPLIT

What if we change the split? Same model, same data.

`random_state = 42`	R^2 0.58
`random_state = 7`	R^2 0.64

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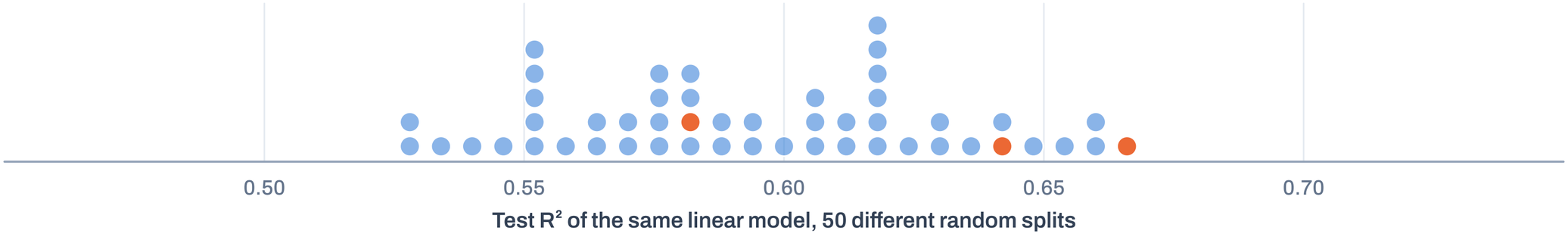
random_state = 42	R^2 0.58
random_state = 7	R^2 0.64
random_state = 21	R^2 0.66

ONE SPLIT IS ONLY ONE SPLIT

What if we change the split? Same model, same data.

random_state = 42	R^2 0.58
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Test R^2 from 50 different random splits: 0.53 to 0.66

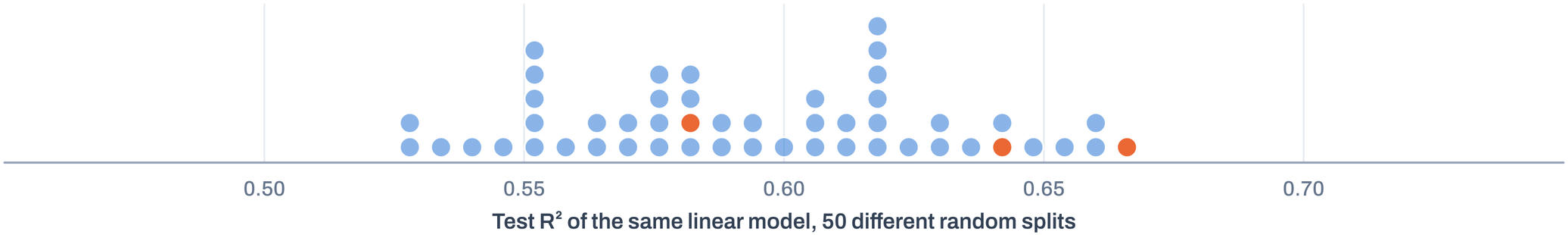


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Test R^2 from 50 different random splits: 0.53 to 0.66



Which mixes land in the test set changes the score. A single split gives one estimate of performance.

A MORE SYSTEMATIC VIEW

K-fold cross-validation

1,005 mixes → 80% training: 804 | 20% test: locked

5-FOLD CV ON THE TRAINING DATA ONLY

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Fold 1	validate	train	train	train	train

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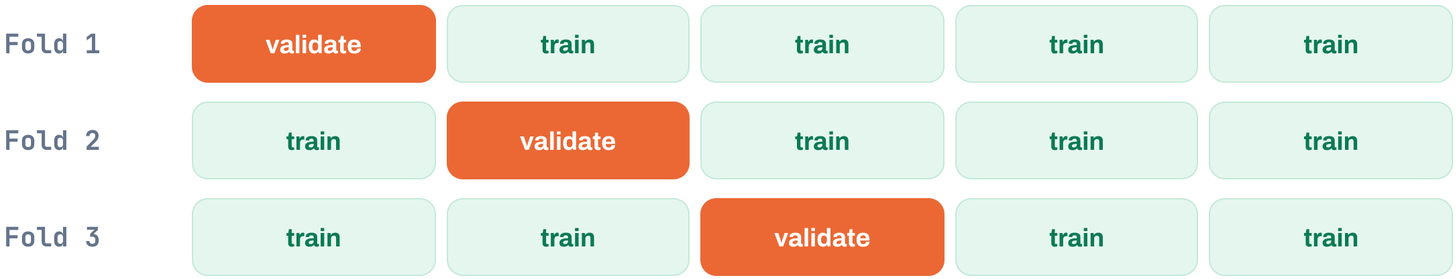
Fold 1	validate	train	train	train	train
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Average performance metric = $(1/K)\ \Sigma\ score_k$

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EACH MIX GETS ONE TURN

Every training observation is validated exactly once, by a model that did not see it.

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Session03_first_model.ipynb — cv

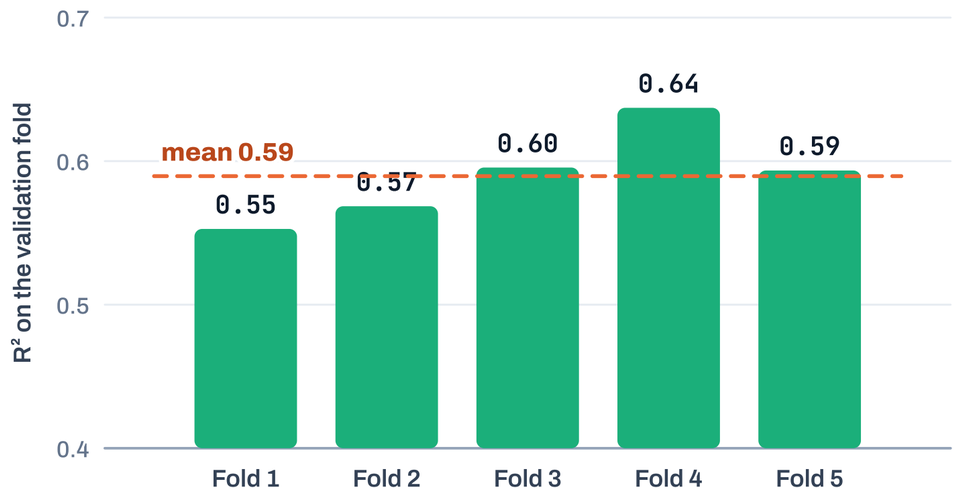
```
from sklearn.model_selection import \
    cross_val_score, KFold
cv = KFold(5, shuffle=True, random_state=42)
scores = cross_val_score(model, X_train,
    y_train, cv=cv, scoring="r2")
```

WHAT DOES CROSS-VALIDATION TELL US? · LOOK BEYOND THE MEAN

Average and spread both matter

Our linear regression: R^2 0.59 ± 0.03

Five validation folds of the training data.

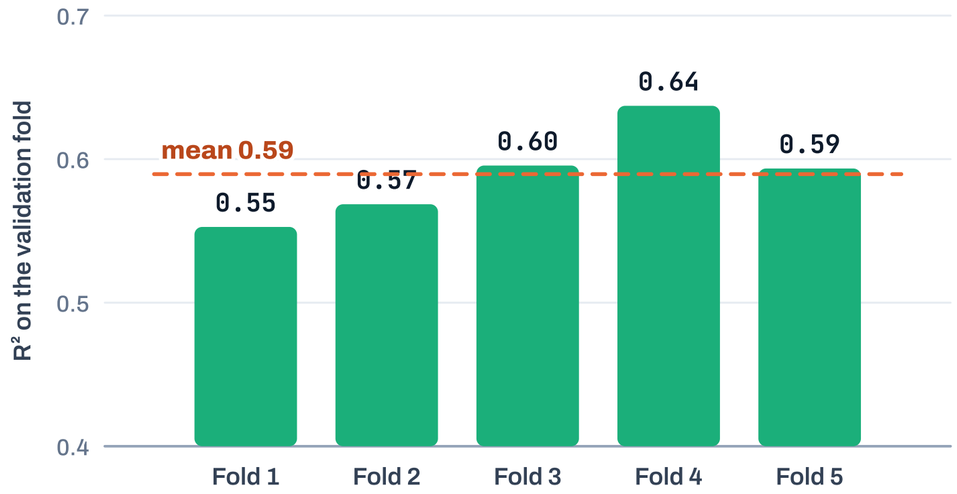


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An illustration with two hypothetical models:

Model A — mean ≈ 0.90 · consistent

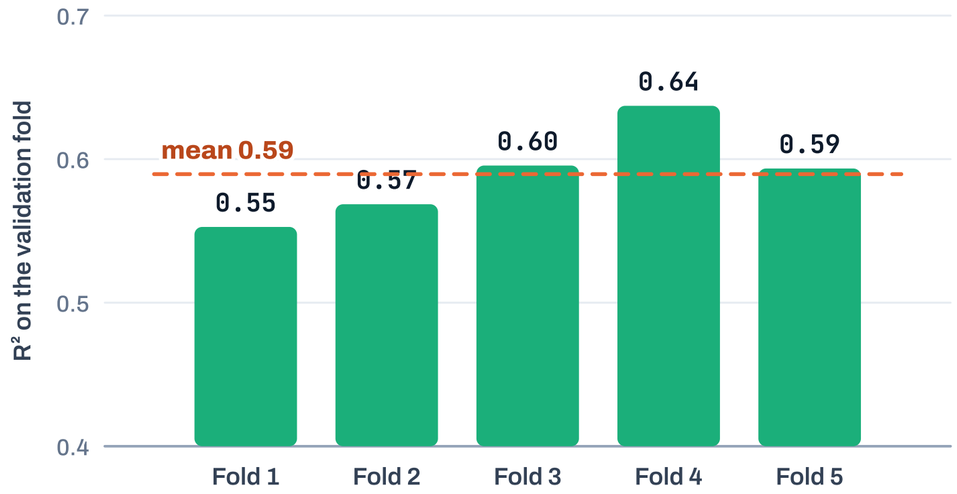
0.89	0.91	0.90	0.92	0.88

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Model B — mean ≈ 0.90 · much more variable

0.60	0.97	0.91	1.00	1.02*

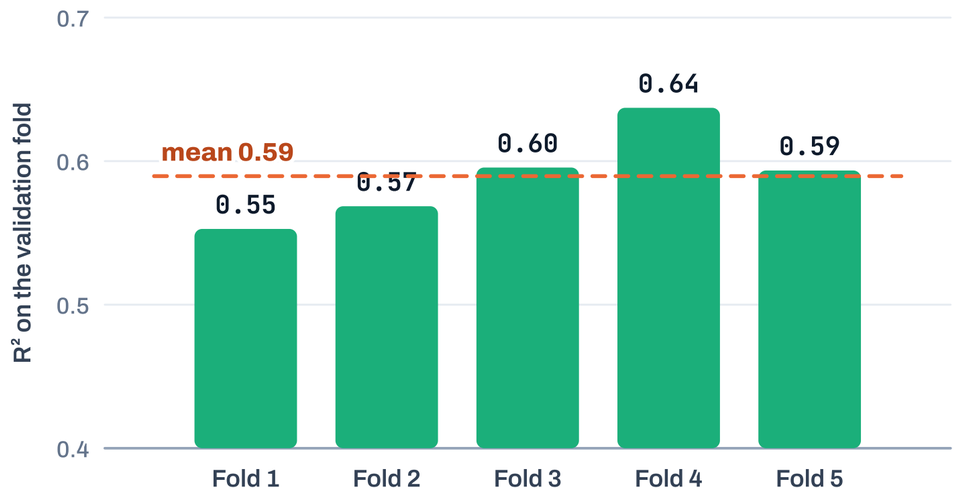
* An R^2 above 1 is impossible: a value like this signals a bug in the evaluation.

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* An R^2 above 1 is impossible: a value like this signals a bug in the evaluation.

Similar average, much larger fold-to-fold variation: Model B deserves more investigation.

THE MOST IMPORTANT RULE FROM TODAY

Keep the test set untouched during development



The test set is for the final evaluation.

Use it once, at the end.

Do not use test performance to decide which

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- **Pre-processing approach**

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is better. Every peek turns the test set into training data.

SEPARATE DEVELOPMENT FROM FINAL ASSESSMENT

The better workflow

01 Training data

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02 Cross-validation

03 Model development and comparison

04 Choose the final approach

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07 Report the final test performance

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Use cross-validation to make decisions.

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Use the test set to make the final assessment.

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↓

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↓

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DECISIONS

Use cross-validation to make decisions.

ASSESSMENT

Use the test set to make the final assessment.

This becomes increasingly important when we compare several algorithms in Sessions 04–06.

FROM MODEL PERFORMANCE TO ENGINEERING JUDGEMENT

Accuracy is only part of the story

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Does it generalise?

Was it evaluated on unseen data?

Test R^2 0.58, CV 0.59

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Is the error acceptable?

What do MAE and RMSE mean in engineering terms?

±8.9 MPa on average

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Does the model produce sensible results?

Straight-line age effect: no

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Are errors concentrated in certain regions?

Early ages, high strength

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Are we extrapolating?

Is the new mix outside the training range?

Check against df.describe()

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Check against df.describe()

Machine learning gives us a model. Engineering judgement tells us whether to trust it.

LIVE EXERCISE · IN THE NOTEBOOK

Your turn: diagnose the model

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A

Change the random seed

Does the test performance change? By how much?

```
random_state = 7, 21, …
```

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Baseline vs model

Did linear regression actually improve on the mean prediction? On which metric, by how much?

```
DummyRegressor()
```

LIVE EXERCISE · IN THE NOTEBOOK

Your turn: diagnose the model

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Inspect the residuals

- Centred around zero?
- Any curvature?
- Does the spread change?
- Unusually large residuals?

```
plt.scatter(X_test.age, res)
```

LIVE EXERCISE · IN THE NOTEBOOK

Your turn: diagnose the model

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ENGINEERING QUESTION **Is there evidence that our linear model is too simple for this problem?**

WHAT WE LEARNED TODAY · FROM ENGINEERING DATA TO A VALIDATED PREDICTION

The complete first modelling workflow

WHAT WE LEARNED TODAY · FROM ENGINEERING DATA TO A VALIDATED PREDICTION

The complete first modelling workflow

01

Engineering question

02

Features + target

03

Train / test split

04

Baseline

05

Linear regression

06

Predictions

WHAT WE LEARNED TODAY · FROM ENGINEERING DATA TO A VALIDATED PREDICTION

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06
Predictions

07
MAE · RMSE · R^2

08
Residual diagnostics

09
Split sensitivity

10
Cross-validation

11
Final test assessment

12
Engineering judgement

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The complete first modelling workflow

01 **Engineering question**

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05 **Linear regression**

06 **Predictions**

07 **MAE · RMSE · R^2**

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09 **Split sensitivity**

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12 **Engineering judgement**

THE CENTRAL LESSON

A prediction is not enough.

We need to know how well it works, how consistently it works, and whether the result makes engineering sense.

WHERE DO WE GO NEXT?

What if the relationship is not linear?

TODAY'S EVIDENCE

Linear regression assumes a simple, straight-line relationship. Our residuals curve with age. Civil engineering problems are often:

- Nonlinear
- Interactive
- Threshold-dependent
- Highly coupled

So what happens when a straight-line model is not enough?

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So what happens when a straight-line model is not enough?

NEXT · SESSION 04

Trees & ensemble models

Decision trees → Random Forest → Boosting

Same engineering problem. Same data.

A different modelling approach.

Compared properly, with today's workflow.

BEFORE SESSION 04

Three small things to try

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1

Ten seeds

Repeat the split with ten random seeds. Record the test MAE each time: what is the range?

```
for s in range(10): …
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BEFORE SESSION 04

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Add log(age)

Add a feature `log(age)`. Does cross-validated R^2 improve? Do the residuals still curve?

```
np.log(df["age"])
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BEFORE SESSION 04

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Your acceptable error

For one decision in your field, write the error you could accept. Compare it with today's MAE.

```
MAE ≤ ? MPa
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BEFORE SESSION 04

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04 **Nonlinear Engineering Problems: Trees & Ensembles**

Decision trees, Random Forest and boosting on the same data.

Questions

Unmute, or drop your question in the chat.



Linear regression

Gradient descent

Metrics & residuals

Cross-validation

Dr. Abdul Waris

Research Associate · Universität Siegen, Germany

END OF SESSION 03

Thank you.

See you in session 04: **Nonlinear Engineering Problems: Trees & Ensembles.**

Dr. Abdul Waris
Research Associate · Universität Siegen, Germany

Course page
qudeltastudios.com/courses

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APPENDIX · FOR THE INSTRUCTOR

Recommended timing for the 90-minute session

PART	SLIDES	APPROX. TIME
Problem, dataset, baseline, training, linear regression	1–14	12 min
Notebook load · drop duplicates · split · baseline · fit	—	15 min
The maths: cost function, gradient descent, learning rate, scaling	15–19	10 min
Metrics and train vs test	21–26	8 min
Notebook predictions · MAE · RMSE · R^2 · parity and residual plots	27–28	15 min
Split sensitivity, cross-validation, test-set rule, workflow	30–34	10 min
Notebook cross-validation · live exercise	36	12 min
Engineering judgement, summary, next session, questions	35, 37–41	8 min
Total		90 min